



Frontal dynamics and phytoplankton decoupling in the Alborán Sea from high-resolution satellite data

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ABSTRACT

The Alborán Sea, in the westernmost Mediterranean, is dominated by oceanic fronts that sustain high primary production through strong physical–biological coupling. Whether this linkage is changing under current warming trends remains poorly understood. Here, we analyse the seasonal and interannual variability of sea surface temperature (SST, 2008–2024) and chlorophyll-*a* (chl-*a*, 1998–2024) fronts using 1-km resolution satellite data. Applying the Canny Edge Detector algorithm, we derived two complementary metrics: frontal gradient (i.e., sharpness) and frontal probability (i.e., detection frequency). SST-derived frontal metrics exhibit a consistent seasonal cycle and have intensified over time, whereas chl-*a* concentration and frontal features show more variable seasonal patterns and long-term declines. This divergence indicates a progressive decoupling between physical frontal structures and phytoplankton response in the Alborán Sea. We suggest that increasing stratification and shifts in bloom phenology may limit the capacity of surface fronts to sustain phytoplankton growth, even where frontal activity persists. These findings provide early evidence of ecosystem vulnerability and underscore the importance of analysing physical and biological dynamics together at fine scales.

1. Introduction

The rise in sea surface temperature (SST) due to climate warming is reshaping upper-ocean physical processes, including turbulent mixing, stratification and frontal dynamics, with major implications for nutrient supply and plankton responses (e.g., Behrenfeld et al., 2006; Mahadevan, 2016; Lévy et al., 2018; Ruiz et al., 2019). These modifications in upper-ocean structure affect the pathways through which energy and nutrients are redistributed, ultimately shaping the spatial organisation of marine ecosystems. Oceanic fronts, which are critical for vertical exchange in the upper water column (e.g., Rudnick, 1996; Mahadevan, 2016; Rousselet et al., 2019), are of particular interest. These fronts are narrow zones that separate water masses with different vertical structures, and are characterised by sharp horizontal gradients of water properties (Le Fèvre, 1987; Belkin, 2002). They occur across a wide range of spatial and temporal scales — from the submesoscale (~0.1–10 km, hours to days) to the mesoscale (~10–100 km, weeks to months) — and can form through various physical mechanisms, including wind-driven upwelling (e.g., Capet et al., 2008), tidal mixing (e.g., Simpson and Hunter, 1974), internal waves (e.g., Gregg, 1987), river

plumes (e.g., Horner-Devine et al., 2015) and mesoscale eddies (e.g., McGillicuddy et al., 1998).

Oceanic fronts play a crucial role in shaping marine ecosystems (Le Fèvre, 1987; Belkin, 2002; Longhurst, 2007; Olson et al., 2015), as they are associated with increased biological activity due to the lateral density gradients they create. On the denser side of the front, positive (cyclonic) vorticity is produced, while negative (anticyclonic) vorticity develops on the less dense side (Mahadevan, 2016). To restore geostrophic balance, a secondary cross-front ageostrophic circulation is induced, leading to the upwelling of nutrient-rich waters into the sunlit layers on the anticyclonic side (Mahadevan and Archer, 2000; Lévy et al., 2001; Rodríguez et al., 2001; Lévy, 2003), and the subduction of organic matter and oxygen to deeper layers on the cyclonic side (e.g., Spall and Richards, 2000; Lévy et al., 2001; Rivière and Pondaven, 2006; Mahadevan et al., 2012; Omand et al., 2015).

Frontal dynamics are central to the high productivity of the Alborán Sea (Oguz et al., 2016), the westernmost basin of the Mediterranean (Fig. 1). The inflow of relatively fresh Atlantic Water through the

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Strait of Gibraltar, known as the Atlantic Jet (AJ), generates a wave-like frontal system characterised by strong lateral density gradients between saltier Mediterranean Water to the north and fresher modified Atlantic Water to the south (e.g., Arnone et al., 1990; La Violette et al., 1990; Tintoré et al., 1991; Viúdez et al., 1996; Millot, 1999; Vélez-Belchí et al., 2005). The AJ feeds two large anticyclonic gyres: the Western Alborán Gyre (WAG), between the Strait of Gibraltar and 3.50° W, and the Eastern Alborán Gyre (EAG), between ~2.50° and 1.50° W (e.g., Crepon, 1965; Lanoix, 1974; Viúdez et al., 1996; Renault et al., 2012). Along their frontal boundaries, strong ageostrophic secondary circulation induces vertical velocities up to $\sim 6 \cdot 10^{-4} \text{ m s}^{-1}$ (García-Jové et al., 2022). In addition, the AJ drives two cyclonic features that further shape the basin circulation: the Central Cyclonic Gyre (CCG), located between the WAG and EAG, and the Coastal Cyclonic Gyre (CsCG), near the Spanish coast, whose size varies with the extent of the WAG (Sarhan et al., 2000; Macías et al., 2008; Bolado-Penagos et al., 2021). The eastern boundary of this circulation system is set by the Almería–Orán Front (Tintoré et al., 1988; Allen et al., 2001; Zarokanellos et al., 2022), which acts as a barrier to the eastward progression of modified Atlantic Water. These physical structures strongly influence biogeochemical dynamics. Nutrient enrichment along the frontal systems supports elevated phytoplankton biomass and chlorophyll concentrations. However, the intensity of frontal activity and associated nutrient supply decreases eastward, as the AJ weakens with distance from the Strait of Gibraltar (Sala et al., 2022). Despite this ecological significance, the Alborán Sea remains understudied in terms of long-term, fine-scale frontal dynamics and their biological implications, which motivates the present analysis.

The Alborán Sea, a frontal transition zone between Atlantic and Mediterranean waters, provides an ideal setting to examine these processes. This study aims to analyse the seasonal and interannual variability of SST and chlorophyll-*a* (chl-*a*) fronts. Previous studies have explored various aspects of SST and chl-*a* variability in the region, particularly in relation to dominant hydrodynamic structures such as the AJ–WAG system (e.g., Falcini et al., 2015; Pascual et al., 2017; Abdellaoui et al., 2024). However, the role of mesoscale and submesoscale frontal structures (i.e., their frequency, intensity, and seasonal timing) has not been fully characterised, especially at resolutions capable of capturing their short-term variability and ecological relevance. Here, we investigate whether the historically tight coupling between surface fronts and phytoplankton biomass is shifting over time. We hypothesise that, under progressive warming, frontal structures may persist or intensify due to enhanced thermal gradients, yet their biological effectiveness may decline due to reduced nutrient supply or altered bloom timing; that is, a physical–biological decoupling with potential consequences for ecosystem productivity and resilience.

Our goal is to provide the first long-term, high-resolution assessment of frontal dynamics and their ecosystem impacts in this rapidly changing transition zone. To this end, we analysed 17 years of satellite-derived SST data and 27 years of chl-*a* data at 1-km spatial resolution, applying the Canny Edge Detector algorithm to identify and quantify surface frontal features. While previous efforts have detected fronts in the Alborán Sea using alternative methods, this study offers the first consistent, fine-scale characterisation of their seasonal variability and interannual trends, alongside associated biological responses. By computing frontal gradient and probability metrics from high-resolution satellite imagery, we reveal patterns in physical–biological coupling with high temporal and spatial detail.

2. Data and methodology

2.1. Satellite images

2.1.1. Sea surface temperature imagery

The SST (°C) images used in this study were downloaded from Copernicus Marine Environment Monitoring Service (CMEMS, [http://](http://marine.copernicus.eu/)

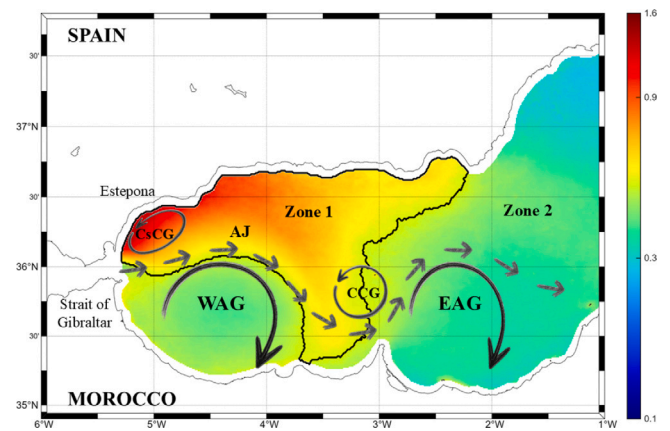


Fig. 1. Map of the Alborán Sea showing the main surface circulation features, overlaid on the mean chlorophyll-*a* concentration (mg m^{-3}) for the entire study period (1998–2024). The black solid line delineates the mesotrophic subregion (Zone 1), while the remaining area corresponds to the oligotrophic subregion (Zone 2); both Zones are defined in Section 3.3. Major surface circulation structures include the Atlantic Jet (AJ), Western Alborán Gyre (WAG), Eastern Alborán Gyre (EAG), Central Cyclonic Gyre (CCG) and the Coastal Cyclonic Gyre (CsCG).

marine.copernicus.eu/; Buongiorno Nardelli et al., 2013). The Mediterranean Sea - High Resolution and Ultra High Resolution L3S Sea Surface Temperature product provides daily merged SST data from multiple sensors, retaining only the highest-quality pixels from Level-2 Preprocessed (L2P) images, which were then remapped over the Mediterranean Sea at $\sim 1\text{-km}$ (0.01°) spatial resolution. All available images from January 1st 2008 to December 31st 2024 were downloaded considering a region of interest centred in the Alborán Sea, between 6.00–1.00° W and 34.90–37.80° N (see Fig. 1). A total 6204 daily images were downloaded.

2.1.2. Ocean colour imagery

The satellite ocean colour data used in this study were downloaded from the GlobColour Project (<https://www.globcolour.info/>). We used the CHL1 (surface chl-*a*, mg m^{-3}) merged product. This merged product was computed with the GSM (Garver-Siegel-Maritorena) model, which merges single sensor Level-3 (L3) normalised remote sensing reflectance (nRrs) at the original sensor wavelengths, without intercalibration (Maritorena and Siegel, 2005). Several sensors, such as SeaWiFS (Sea-viewing Wide Field-of-view Sensor), MODIS (MODerate resolution Imaging Spectroradiometer), MERIS (MEDium Resolution Imaging Spectrometer), VIIRS (Visible Infrared Imaging Radiometer Suite) and OLCI (Ocean and Land Colour Instrument), are considered. The final product consists of daily averages at $\sim 1\text{-km}$ (0.01°) resolution already remapped to a Mercator projection. All available images from January 1st 1998 to December 31st 2024 were downloaded via FTP (File Transfer Protocol) considering the same region of interest as in the SST imagery. A total 9626 daily images were downloaded.

2.2. Canny Edge Detector algorithm

To detect frontal regions in our area of interest, we applied the Canny Edge Detector (CED) algorithm (Canny, 1986) to both SST and chl-*a* datasets. The CED algorithm has been successfully applied previously for the detection of oceanic fronts using SST (e.g., Castelao et al., 2005) and ocean colour imagery (e.g., Royer et al., 2004). To implement the CED algorithm, four steps were followed (see Fig. 2).

Step 1 involves smoothing the images to remove residual noise at the pixel level, achieved by convolution with a Gaussian kernel filter

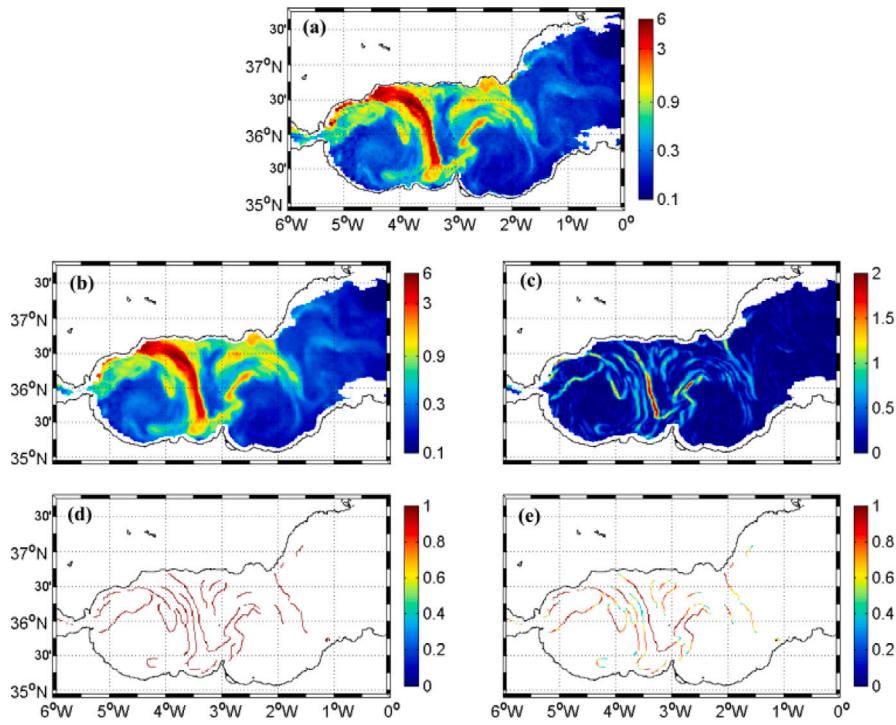


Fig. 2. Case study: May 17th, 2005. Example of the sequential steps to implement the Canny Edge Detector (CED) algorithm. (a) Original chlorophyll-*a* image (mg m^{-3}). (b) Step 1: Smoothing of the image. (c) Step 2: Magnitude of the gradient. (d) Step 3: Non-maximal suppression. (e) Step 4: Identification of frontal pixels.

(Eq. (1), see Fig. 2b):

$$G(x, y) = \frac{1}{2\pi\sigma} \cdot e^{-\frac{x^2+y^2}{2\sigma^2}}, \quad (1)$$

where x and y denote absolute pixel coordinates, and σ is the standard deviation (width) of the distribution. A 5×5 Gaussian kernel filter was applied.

Next, to settle an optimal σ value, we followed the equation defined by Oram et al. (2008) (Eq. (2)):

$$\sigma = \frac{\text{edge scale (km)}}{\text{image resolution (km pixel}^{-1}\text{)}}, \quad (2)$$

An edge scale of 3 km was selected because it suppresses pixel-scale noise while preserving the fine-scale (1–10 km) frontal gradients characteristic of the Alborán Sea. Selecting a desired edge scale of 3 km, with an image resolution of 1 km, the optimal σ value is 3. This σ value was applied uniformly to all images, ensuring consistent detection of frontal gradients across the entire dataset, and the same configuration was used for both SST and chl-*a* fields.

The objective of **Step 2** is to obtain the magnitude (Eq. (3)) and orientation (Eq. (5)) of the gradient (Fig. 2c). First, the gradient magnitude (G_m) is obtained applying the first derivative of the Sobel operator:

$$G_m = \sqrt{G_x^2 + G_y^2}, \quad (3)$$

with G_x defining the image gradient along the x direction and G_y along the y direction (Eq. (4)). A simple way to compute G_x and G_y is to use the convolutional masks (Belkin and O'Reilly, 2009; Nieto, 2009):

$$G_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 1 & 2 \\ -1 & 0 & 1 \end{pmatrix}, \quad G_y = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix} \quad (4)$$

Second, the gradient direction (G_d) is computed as:

$$G_d = \tan^{-1} \frac{G_y}{G_x}, \quad (5)$$

Step 3 is known as Non-Maximal Suppression. Its purpose is to detect and delete false edges. Only those pixels at which the gradient

magnitude is locally maximum along the gradient direction are considered edge pixels (Nieto, 2009) (Fig. 2d). Hence, the algorithm identifies all the pixels that do not belong to frontal edges and assigns them a zero value.

Finally, **Step 4** aims to set up a minimum (T_{min}) and a maximum (T_{max}) threshold value to properly select the frontal pixels and suppress the hysteresis effect that may arise from the previous step (Fig. 2e). Here, we followed Oram et al. (2008), who established T_{min} and T_{max} thresholds at the 80th and 90th percentiles, respectively, based on a global histogram of the normalised gradient magnitude.

In this study, the CED algorithm was applied to each daily SST and chl-*a* image. In the case of chl-*a* concentration data, and given that chlorophyll concentration follows a log-normal distribution (Campbell, 1995), the original chl-*a* data were log-transformed (Belkin and O'Reilly, 2009) before applying the algorithm.

In addition to the gradient (G_m) of the frontal pixels (herein Frontal Gradient; FG , units km^{-1}), we will also analyse the frontal probability (Belkin et al., 2009). Frontal Probability (FP , %; Eq. (6)) determines the frequency of a pixel being classified as a front during a defined period as:

$$FP = \frac{N}{C} \cdot 100, \quad (6)$$

where N indicates the number of images in which a pixel was detected as a front, and C indicates the number of images in which the same pixel was cloud-free. FP values of 0% mean that the pixel was never identified as a front, while values of 100% mean that the pixel was always identified as a frontal pixel during that period. It should be noted that FP decreases as the studied period increases (i.e., FP will show higher values at lower temporal scales).

This analysis yields four frontal metrics that will serve as key descriptors in our study: the magnitude of SST and chl-*a* frontal gradients (FG_{sst} and FG_{chl}), and the probability of frontal occurrence based on each variable (FP_{sst} and FP_{chl}). Together, these datasets offer complementary perspectives on frontal activity, capturing both its intensity and persistence across space and time.

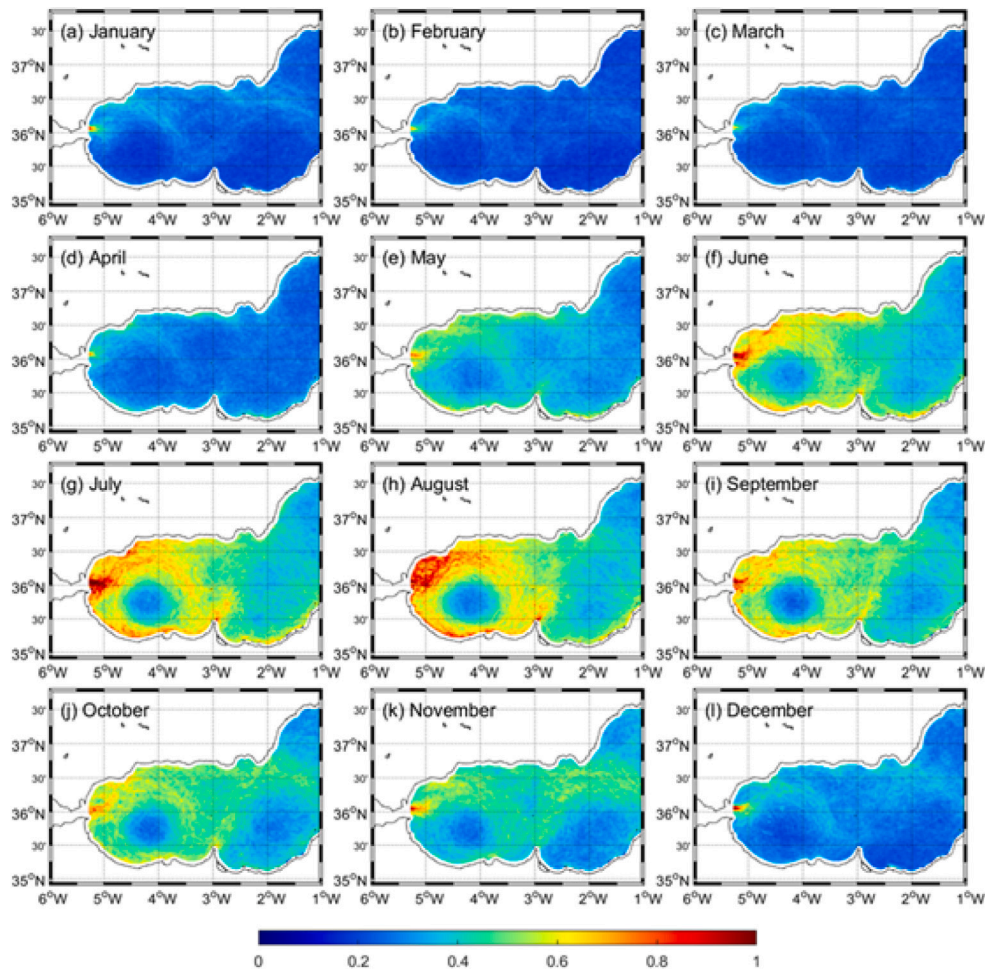


Fig. 3. Monthly climatology of sea surface temperature frontal gradient (FG_{sst} , $^{\circ}\text{C km}^{-1}$) for the 2008–2024 period.

2.3. Data treatment

The region of the Alborán Sea is characterised by frequent cloudy conditions resulting in patchy spatial coverage and temporal gaps. To avoid this problem, composite images were computed by averaging pixel-by-pixel the values of the corresponding daily images. Furthermore, to reduce the influence of noisy coastal pixels in the analyses, pixels within ~ 5 km of the shoreline were manually masked out.

We computed two composite datasets for all the variables (SST , FG_{sst} , FP_{sst} , $chl-a$, FG_{chl} , and FP_{chl}). A weekly (7-day) composite dataset was computed to study the interannual and seasonal trends, since it offers high sensitivity to the short-term and fine-scale processes that characterise this complex hydrodynamic region. For the SST data, and the corresponding frontal gradient and probability, were generated 888 weekly composite images, all valid images. For the $chl-a$ data, and the corresponding frontal gradient and probability, 1409 weekly composite images were generated, of which 1384 were valid with cloud coverage below 20% of sea pixels; only $\sim 1.77\%$ of the images exceeded this threshold. Weekly composites exceeding 20% cloud coverage were excluded from the dataset. For analyses requiring continuous time series (see below), the resulting short gaps were filled by linear interpolation. The second composite datasets are the monthly climatologies (i.e., 12 images) to study the spatial variability at a seasonal level.

2.4. Wavelet analysis

We performed a Wavelet Power Spectrum (WPS) analysis to examine the temporal changes and periodicities of the weekly-averaged time

series for all the variables (SST , FG_{sst} , FP_{sst} , $chl-a$, FG_{chl} , and FP_{chl}), which are crucial for understanding the dynamics of the study area. Due to the method's requirement for uninterrupted datasets, missing time steps were filled by linear interpolation.

Over the past decade, wavelet analyses have been effectively applied to ecological non-stationary time series, enabling local time-scale decomposition of signals in both the time and frequency domains (e.g., Lau and Weng, 1995). The WPS illustrates the relative frequency contributions at each time step on a time-frequency plane (Cazelles et al., 2008). To mitigate false periodicities, data points near the series edges, marked by the cone of influence, were excluded, with caution exercised in interpreting data below this boundary (Torrence and Compo, 1998). Additionally, to assess the non-random nature of wavelet-based metrics, a 5% significance level was established using a bootstrapping approach with a hidden Markov model (Cazelles and Stone, 2003), implemented using the MATLAB toolbox developed by Cazelles and Chavez (Cazelles et al., 2008). The Morlet wavelet was selected for its optimal balance between time and frequency localisation (Grinsted et al., 2004).

2.5. Quantile regression analysis

To explore long-term trends in the weekly-averaged time series of SST , $chl-a$, and their associated frontal metrics, we applied Quantile Regression (QR) following Wang et al. (2023). QR estimates trends across different parts of a distribution, capturing asymmetries and changes in variability not detected by mean-based approaches. We fitted linear models at the 0.25th, 0.50th (median), and 0.75th quantiles,

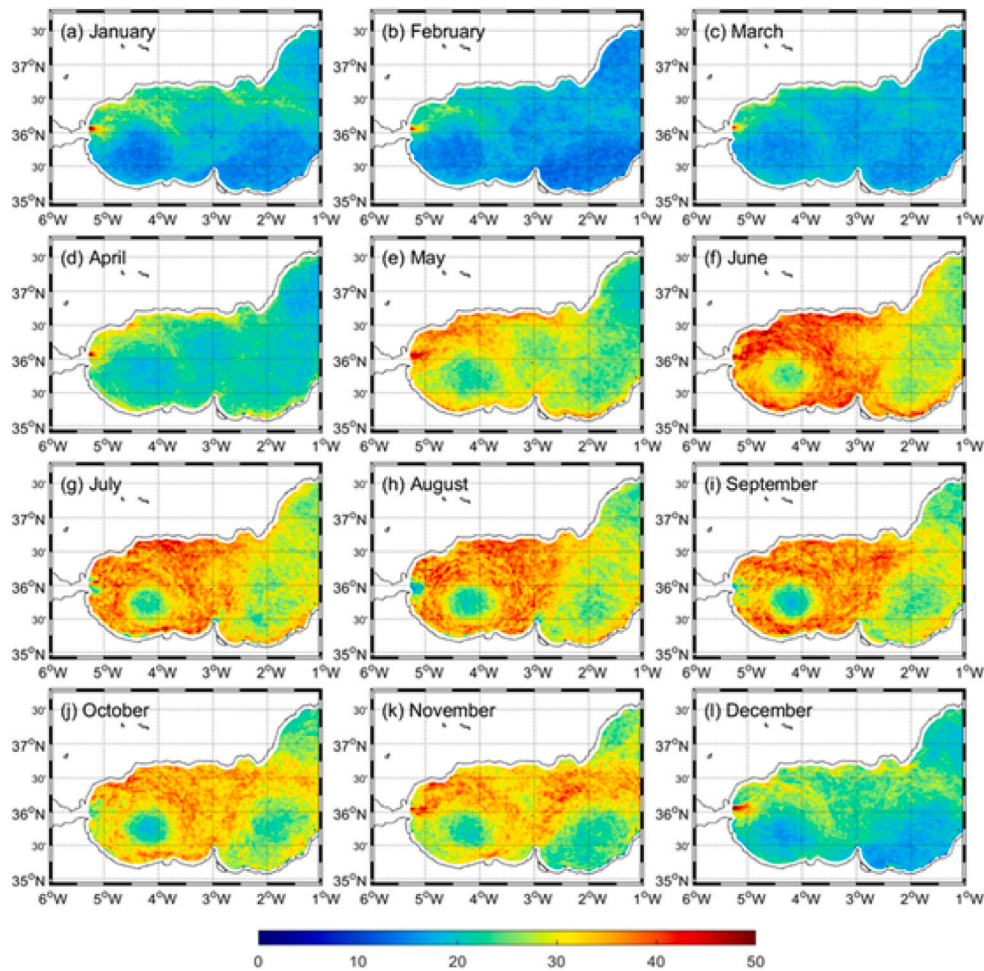


Fig. 4. Monthly climatology of sea surface temperature frontal probability (FP_{sst} , %) for the 2008–2024 period.

corresponding to trends in the lower, central, and upper parts of the data range.

The analysis was performed using the *Quantile.reg* toolbox in MATLAB (Wang et al., 2023). Time was expressed in decimal years, and missing values were filled by linear interpolation. For each quantile, we estimated the slope and its standard deviation, and the intercept of the fitted line, along with a 95% confidence interval for the slope derived from 500 bootstrap resamples. Trends were considered statistically significant when the confidence interval did not include zero.

3. Results

3.1. Seasonal variability of sea surface temperature frontal gradient and probability

To simplify the results, we focused only on the description of the frontal variables (i.e., FG_{sst} and FP_{sst}), as the seasonality of SST and the associated frontal gradient and probability, are closely related. Figs. 3 and 4 show the monthly climatologies, computed for the 2008–2024 period, of FG_{sst} and FP_{sst} , respectively, along the Alborán Sea.

SST shows a clear seasonal variability. FG_{sst} analysis reveals the colder-than-Mediterranean AJ signal bordering the WAG throughout the year (Fig. 3). In most months (e.g., January, February, and September–December), it is also observed bordering the EAG. During the winter months (January–March), FG_{sst} ranges from approximately $0.15\text{ }^{\circ}\text{C km}^{-1}$ in offshore regions and $0.50\text{ }^{\circ}\text{C km}^{-1}$ in areas associated with the AJ–WAG front (and with the AJ–EAG front when present). From May onwards, the temperature gradient between the AJ and

the warmer Mediterranean waters increases, reaching maximum values between July and August, when FG_{sst} peaks at $1.00\text{ }^{\circ}\text{C km}^{-1}$ in the AJ–WAG frontal region and drops below $0.40\text{ }^{\circ}\text{C km}^{-1}$ in oligotrophic oceanic areas. Subsequently, FG_{sst} progressively decreases through late autumn and early winter (December), returning to values similar to those observed in January.

FP_{sst} shows a similar seasonal pattern to FG_{sst} . As expected, throughout the year, higher FP_{sst} values are associated with frontal regions (AJ–WAG and AJ–EAG), whereas lower values are observed in oligotrophic offshore zones (Fig. 4). From January to March, FP_{sst} ranges from approximately 15% in oceanic regions to peaks of about 30% in frontal and coastal zones. Starting in April, a progressive increase in FP_{sst} is observed, with maximum values in June–July in frontal regions (reaching up to 40%). Between June and September, the number of frontal pixels increases in the western part of the Alborán Sea, covering almost the entire basin except for the centre of the oligotrophic WAG. From September onward, the number of frontal pixels begins to decrease, and in October and November, frontal pixels with FP_{sst} values around 40% are clearly observed surrounding both the WAG and the EAG. In December, both the number of frontal pixels and their FP_{sst} values decrease substantially, returning to levels similar to those observed in January.

3.2. Seasonal variability of chlorophyll-a frontal gradient and probability

Following the same principle to simplify the results, we describe only the monthly variability of chl-a frontal gradient (FG_{chl} ; Fig. 5)

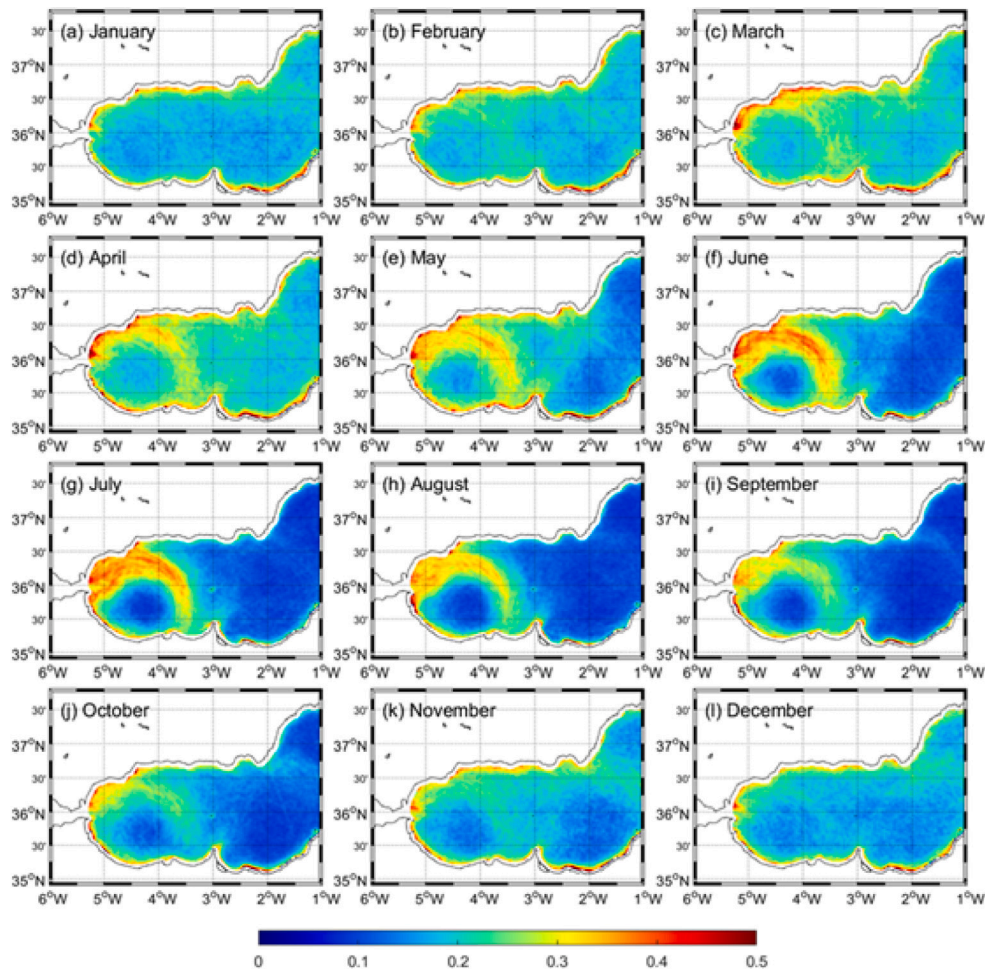


Fig. 5. Monthly climatology of chlorophyll-*a* frontal gradient (FG_{chl} , $\text{mg m}^{-3} \text{ km}^{-1}$) for the 1998–2024 period.

and probability (FP_{chl} ; Fig. 6) for the study period (1998–2024) along the Alborán Sea.

Both variables, FG_{chl} and FP_{chl} , also exhibit a relatively clear seasonal cycle. A progressive increase in FG_{chl} and FP_{chl} is observed from January to June–July, followed by a gradual decrease towards December. In January, FG_{chl} ranges from approximately $0.15 \text{ mg m}^{-3} \text{ km}^{-1}$ in oligotrophic regions to $0.25 \text{ mg m}^{-3} \text{ km}^{-1}$ in coastal regions (Fig. 5), across the entire study area except for some pixels along the coast. While the spatial distribution may appear homogeneous, this apparent uniformity is largely due to the use of a colourmap that is consistent throughout the entire year. Although visually flat, these early-winter maps still reflect meaningful spatial variability (see Fig. A1). As the year progresses into spring and early summer, FG_{chl} decreases in oligotrophic zones to values around $0.05 \text{ mg m}^{-3} \text{ km}^{-1}$, while it increases markedly in the AJ-WAG-CsCG system, where maximum values exceeding $0.40 \text{ mg m}^{-3} \text{ km}^{-1}$ are reached. That is, the difference in chl-*a* concentration between neighbouring pixels in this region increases, reflecting a sharper gradient. From April to August, the frontal intensity strengthens and becomes increasingly confined to the AJ-WAG-CsCG system, where a well-defined high-gradient core develops. Additionally, the seasonal evolution highlights the characteristic west–east decrease in productivity across the region. Although weaker, elevated frontal gradients persist into early autumn around the AJ-WAG-CsCG system, followed by a transition to winter in which the pattern reverses: FG_{chl} decreases in the AJ-WAG-CsCG system and increases again in oligotrophic regions, reflecting a more uniform distribution of chl-*a* concentration across neighbouring pixels.

We can observe a similar pattern for FP_{chl} (Fig. 6). In January, FP_{chl} ranges from approximately 10% in oligotrophic regions to 25%

in coastal regions. The first signs of seasonal increase appear in March, and as we approach spring and early summer, FP_{chl} progressively intensifies in the AJ-WAG-CsCG system. In June and July, this region shows the highest values, with a 40% probability that a pixel associated with this system will be identified as a frontal pixel. During this period, a marked west-to-east contrast develops, as frontal pixels are almost absent in the eastern part of the study region. Although values begin to decline from August onward, FP_{chl} remains elevated in the AJ-WAG-CsCG system until early autumn. As we move into autumn and early winter, the difference between oligotrophic and coastal zones diminishes, and FP_{chl} becomes more (apparently) uniformly distributed across the basin, although the influence of the AJ-WAG-CsCG system persists throughout the year.

3.3. Climatological and interannual variability

The previous results highlight that frontal activity is not evenly distributed across the Alborán Sea, but is instead concentrated in the region shaped by the AJ-WAG-CsCG system (Figs. 3 and 4). Consequently, this subregion shows the highest chl-*a* concentrations (Figs. 5 and 6), indicative of enhanced phytoplankton biomass and productivity. To capture these differences and avoid diluting meaningful patterns through spatial averaging, we divided the study area into two ecologically distinct subregions: (i) the mesotrophic waters associated with the AJ-WAG-CsCG system (hereafter Zone 1), and (ii) the oligotrophic waters occupying the rest of the basin (Zone 2). This classification was based on the long-term average chl-*a* over the 27-year study period, using standard trophic thresholds: chl-*a* < 0.50 mg

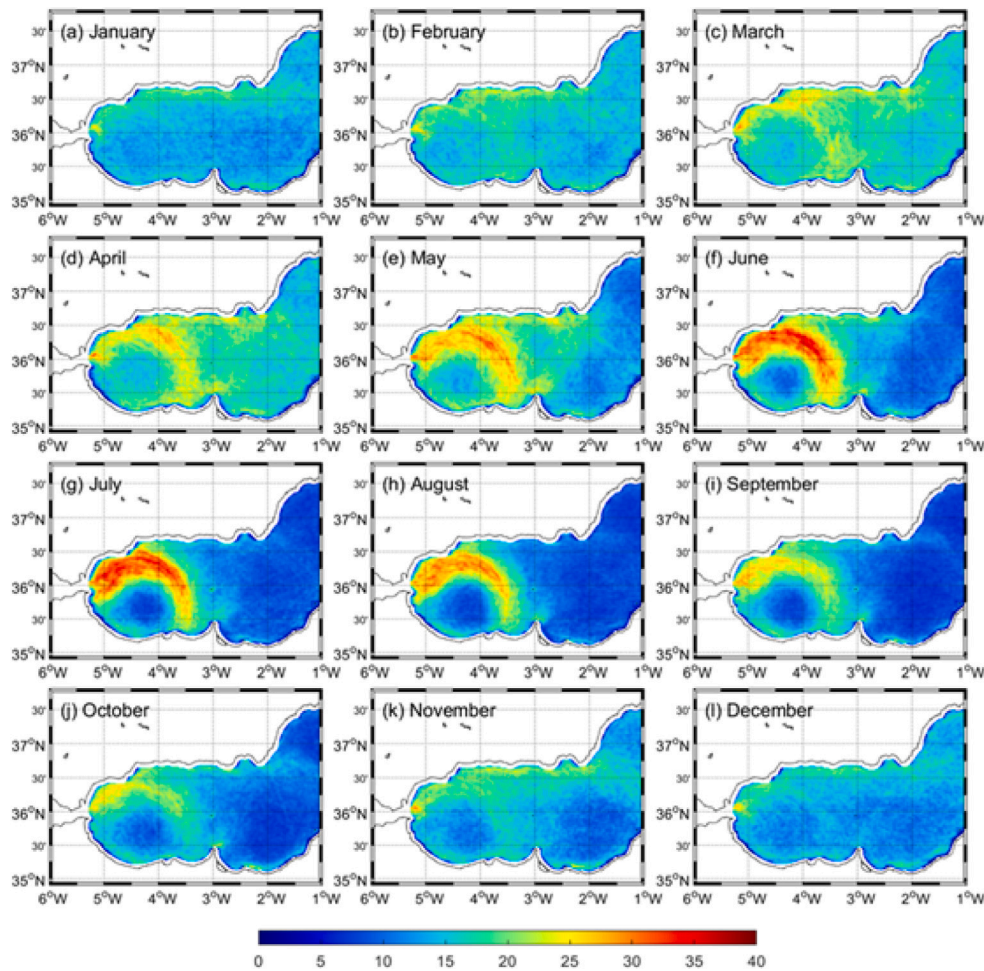


Fig. 6. Monthly climatology of chlorophyll-a frontal probability (FP_{chl} , %) for the 1998–2024 period.

m^{-3} (oligotrophic), $0.50\text{--}1.00 \text{ m}^{-3}$ (mesotrophic), and $\geq 1.00 \text{ mg m}^{-3}$ (eutrophic) (Ignatiades, 2005). The resulting map shows that Zone 1 encompasses mostly mesotrophic waters, with a small eutrophic core near the north-western margin, probably associated with the CsCG (see Fig. 1 and Fig. A2). This zonal approach allows us to assess not only seasonal and interannual patterns but also whether frontal trends differ depending on the underlying trophic regime.

Figs. 7 and 8 show the seasonal variability of SST, chl-*a*, and their associated frontal metrics in the mesotrophic subregion (Zone 1) and the oligotrophic subregion (Zone 2), respectively. The black solid line represents the monthly climatology, and the shaded green area indicates its standard deviation, capturing the interannual variability over the study period. Red dots correspond to weekly averages plotted by day of the year, illustrating the full range of observed values. This visualisation provides an overview of typical seasonal cycles while also capturing the range of interannual variability within each zone. By comparing these patterns across subregions, we establish a baseline understanding of how seasonal dynamics differ between trophic regimes and identify potential asymmetries in intra-annual variability that could precede long-term shifts.

In the mesotrophic subregion (Zone 1, Fig. 7), associated with the AJ-WAG–CsCG system, SST shows a clear seasonal cycle, with weekly values ranging from $\sim 14^\circ\text{C}$ in winter to $\sim 25^\circ\text{C}$ in late summer. The monthly climatology closely reflects this pattern with smoother variability, and the narrow standard deviation band indicates low inter-annual variability (Fig. 7a). FG_{sst} follows a similar seasonal evolution, increasing steadily from winter ($\sim 0.30^\circ\text{C km}^{-1}$) to late summer ($\sim 0.70^\circ\text{C km}^{-1}$), before decreasing again. Variability is moderate, with

weekly averages showing greater dispersion in spring and summer (Fig. 7b). In contrast, FP_{sst} exhibits a broader seasonal cycle, with a pronounced summer minimum ($\sim 65\%$) and a spring maximum ($\sim 85\%$). The standard deviation band indicates higher interannual variability, particularly in late summer (Fig. 7c).

Chl-*a* concentration in this subregion peaks in winter ($\sim 1.20 \text{ mg m}^{-3}$) and declines through spring to a summer minimum ($\sim 0.40 \text{ mg m}^{-3}$). The weekly averages show high dispersion, especially during the bloom period, indicating marked interannual variability in both magnitude and timing (Fig. 7d). FG_{chl} displays a weak seasonal cycle, with values fluctuating between ~ 0.10 and $\sim 0.25 \text{ mg m}^{-3} \text{ km}^{-1}$. The widespread scatter of weekly values throughout the year suggests limited seasonal coherence in frontal strength (Fig. 7e). Despite this, FP_{chl} remains consistently high throughout the year, with values mostly between 80%–95%. The seasonal climatology shows a subtle summer peak around July and August, and the narrow standard deviation band suggests relatively stable frontal persistence over time (Fig. 7f).

In the oligotrophic subregion (Zone 2, Fig. 8), SST follows a clear and smooth seasonal cycle, also ranging from $\sim 14^\circ\text{C}$ in winter to $\sim 25^\circ\text{C}$ in late summer. The monthly climatology closely matches the weekly values, and the narrow standard deviation band reflects very low interannual variability (Fig. 8a). FG_{sst} also shows a marked seasonal cycle, increasing from winter ($\sim 0.30^\circ\text{C km}^{-1}$) to a summer peak ($\sim 0.75^\circ\text{C km}^{-1}$). While the climatology captures this pattern, the weekly averages lie consistently below it, suggesting that the seasonal mean may be influenced by a few years of stronger frontal activity. Still, the standard deviation band remains relatively narrow, indicating limited interannual variability (Fig. 8b). FP_{sst} displays a moderate

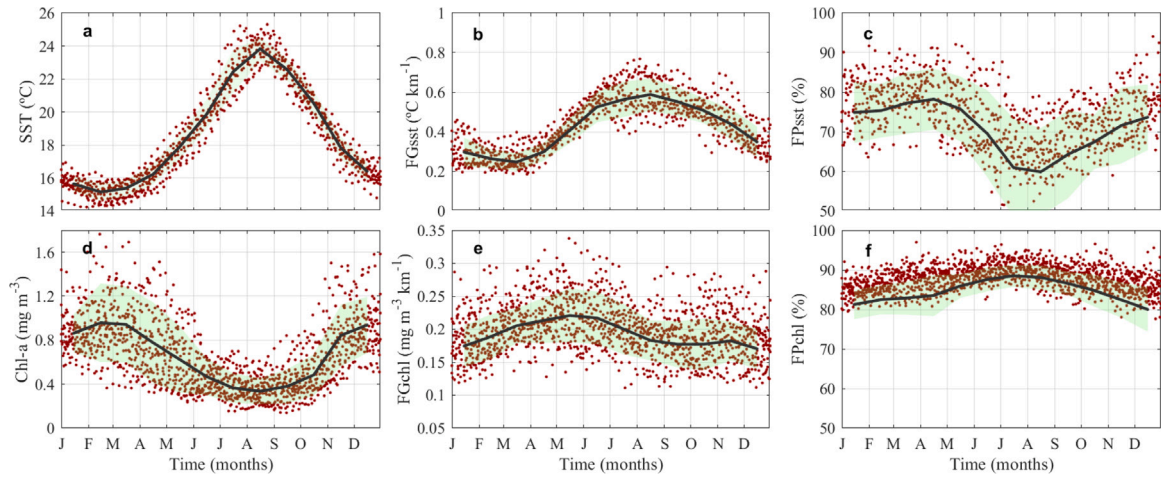


Fig. 7. Seasonal variability in the mesotrophic zone (Zone 1) for: (a) Sea surface temperature (SST, °C), (b) SST frontal gradient (FG_{sst} , °C km⁻¹), (c) SST frontal probability (FP_{sst} , %), (d) chlorophyll-a concentration (chl-a, mg m⁻³), (e) chl-a frontal gradient (FG_{chl} , mg m⁻³ km⁻¹), and (f) chl-a frontal probability (FP_{chl} , %). The black solid line shows the monthly climatology, and the shaded green area represents the standard deviation. Red dots represent weekly averages over the full study period (2008–2024 for SST and its associated frontal metrics; 1998–2024 for chl-a and its associated frontal metrics).

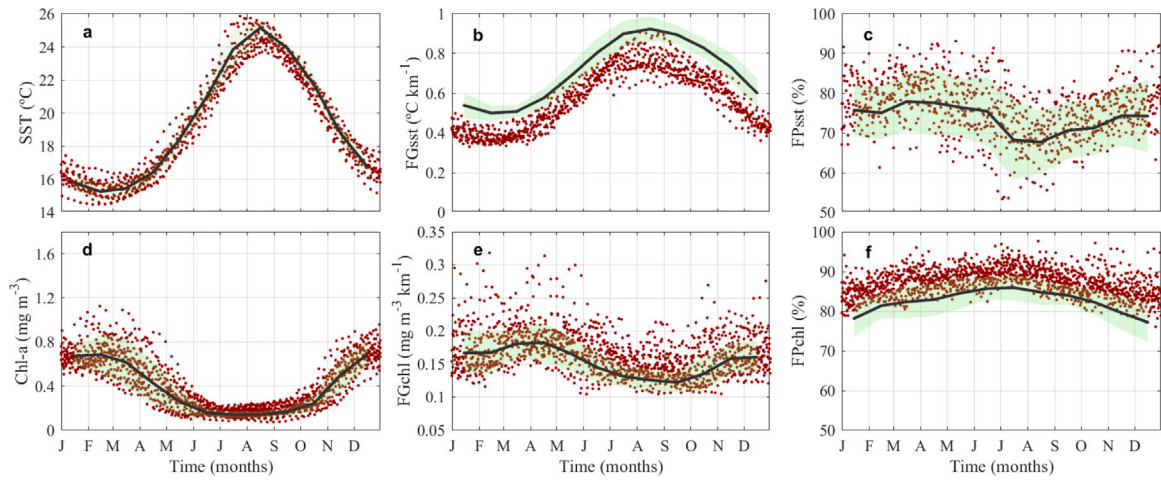


Fig. 8. Seasonal variability in the oligotrophic zone (Zone 2) for: (a) Sea surface temperature (SST, °C), (b) SST frontal gradient (FG_{sst} , °C km⁻¹), (c) SST frontal probability (FP_{sst} , %), (d) chlorophyll-a concentration (chl-a, mg m⁻³), (e) chl-a frontal gradient (FG_{chl} , mg m⁻³ km⁻¹), and (f) chl-a frontal probability (FP_{chl} , %). The black solid line shows the monthly climatology, and the shaded green area represents the standard deviation. Red dots represent weekly averages over the full study period (2008–2024 for SST and its associated frontal metrics; 1998–2024 for chl-a and its associated frontal metrics).

seasonal cycle, with a minimum in summer (~65%) and a maximum in late winter (~85%). The standard deviation band is wider than for SST or FG_{sst} , indicating greater interannual variability (Fig. 8c).

Chl-a concentration peaks in winter (~0.90 mg m⁻³) and declines to a summer minimum (~0.30 mg m⁻³). Weekly values show relatively high dispersion during the bloom period and reduced variability in summer (Fig. 8d). FG_{chl} exhibits a weak seasonal cycle, with values ranging between ~0.10 and ~0.25 mg m⁻³ km⁻¹. The weekly averages are widely scattered throughout the year, indicating limited seasonal coherence in frontal strength (Fig. 8e). FP_{chl} remains high year-round (~80%–95%), with a subtle peak in July. The narrow standard deviation band indicates low interannual variability and a stable frontal structure in this subregion (Fig. 8f).

3.4. Bloom phenology

To analyse potential shifts in the bloom phenology (timing and magnitude) across the mesotrophic and oligotrophic subregions, we examined the seasonal and interannual variability of SST and chl-a concentration using weekly averaged data (red dots in Figs. 7 and 8).

Fig. 9 presents a time-of-year vs. year representation of these two key variables, allowing visual assessment of the consistency and variability in their seasonal cycles. This figure provides a reference framework for comparing not only the timing of seasonal transitions but also the interannual stability of bloom dynamics under differing trophic conditions.

In the mesotrophic subregion, SST follows a clear seasonal cycle, with minimum values (~14 °C) typically occurring between January and March, and maximum values (~25–26 °C) during July–August. While the timing of seasonal transitions remains relatively stable, the figure reveals a gradual warming trend throughout the year, with temperatures appearing slightly higher across all seasons in the later years of the time series (Fig. 9a). Chl-a, by contrast, shows high interannual variability, with no consistent bloom timing across years. Although most blooms occur between January and April, their onset, duration, and intensity vary considerably. Some years exhibit sharp early peaks in January or February, while others show broader or delayed increases extending into spring. Maximum concentrations generally remain around ~1.60 mg m⁻³, but the shape and amplitude of the bloom differ markedly from year to year. During summer, chl-a

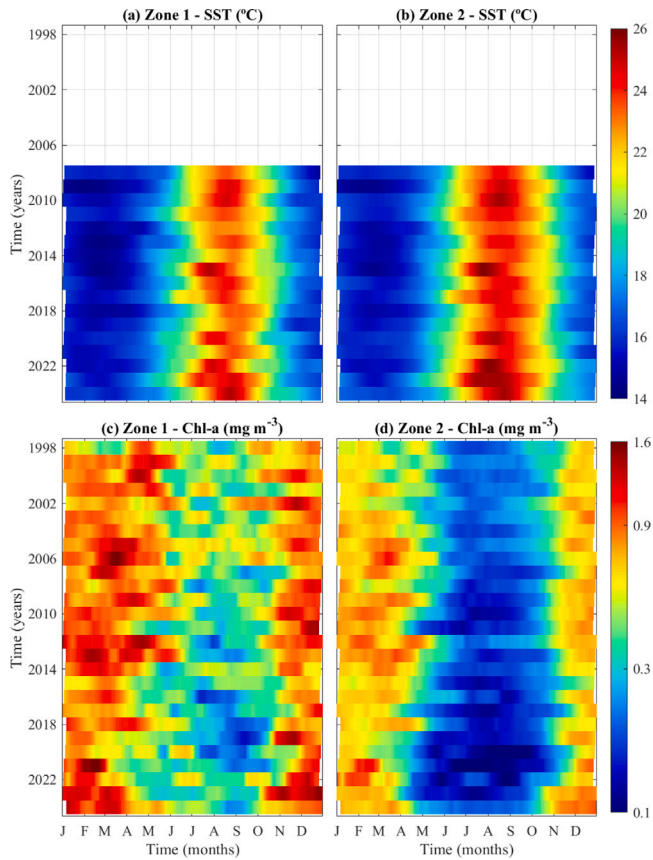


Fig. 9. Bloom phenology in the subregion influenced by the AJ-WAG-CsCG system (Zone 1; left-hand panels) and the oligotrophic waters (Zone 2; right-hand panels), based on weekly averaged data. Panels show the seasonal and interannual variability of (a–b) sea surface temperature (SST, °C) and (c–d) chlorophyll-a concentration (chl-a, mg m⁻³) over the study period (2008–2024 for SST; 1998–2024 for chl-a).

concentrations are consistently low; however, towards the end of the study period, values drop even further, approaching deep oligotrophic conditions (Fig. 9c).

In the oligotrophic subregion, SST also follows a regular seasonal cycle, with minimum temperatures (~14 °C) in winter and maximum values (~25–26 °C) in summer. However, a gradual warming trend is evident over the study period. In recent years, summer temperatures have not only increased but also persisted over a longer period, extending from early June through September (Fig. 9b). Concurrently, chl-a concentrations remain low (<0.90 mg m⁻³) throughout the year but still exhibit a seasonal cycle, although the bloom signal is not visually consistent across years. There is substantial variation in the timing and intensity of seasonal increases: some years show slight rises during winter or spring, while others display a flatter profile. Summer and autumn concentrations are consistently low (~0.30–0.40 mg m⁻³), with little interannual variability. Moreover, a clear decline in chl-a concentration is evident from early summer through autumn in this subregion (Fig. 9d).

3.5. Wavelet power spectrum analyses

To identify the dominant periodicities in both subregions for each time series and to assess how these patterns evolve over time, we performed a WPS analysis. This method allows for the detection of both persistent and intermittent periodicities, providing insight into the temporal dynamics of SST, chl-a and their associated frontal metrics across the study region.

Fig. 10 presents the WPS for SST and chl-a and their frontal metrics in the mesotrophic subregion (Zone 1). SST, FG_{sst} , and FP_{sst} all exhibit a strong annual (1-year) signal throughout the study period (~2019–2024), with slight differences in consistency and intensity. SST shows a robust and statistically significant 1-year periodicity across the entire time series (Fig. 10a). FG_{sst} mirrors this pattern, with the annual signal becoming slightly stronger in the later years (2018–2024; Fig. 10b). FP_{sst} also shows a clear 1-year cycle, though somewhat weaker and more variable over time. Additionally, it displays a low-frequency signal (~5 years), but as most of it lies outside the cone of influence, it cannot be reliably interpreted (Fig. 10c).

In contrast, the chl-a-related metrics (Fig. 10d–f) reveal greater temporal variability. Chl-a exhibits a strong and significant 1-year cycle from ~2003 onward, with the most intense signal occurring after 2010 (Fig. 10d). FG_{chl} shows a weaker and less persistent annual signal, significant only during specific periods (notably 2005–2013 and 2016–2018; Fig. 10e). FP_{chl} maintains a low but statistically significant annual signal across most of the series, except for a coherence gap between 2001 and 2003 (Fig. 10f). Overall, the results suggest that while the thermal front system in this subregion is highly seasonal and consistent, the biological response is more variable, with intermittent coherence at the seasonal scale.

Fig. 11 presents the WPS for SST, chl-a, and their associated frontal metrics in the oligotrophic subregion (Zone 2). The overall patterns resemble those observed in the mesotrophic subregion. For SST and its frontal metrics (FG_{sst} and FP_{sst}), the WPS reveals a strong and statistically significant 1-year signal throughout the study period (~2019–2024; Fig. 11a, b). In the case of FG_{sst} , the annual signal appears slightly stronger during the later years (2018–2024). FP_{sst} (Fig. 11c) also shows a persistent 1-year signal, although weaker and more variable than SST and FG_{sst} . Again, a low-frequency (~5-year) component emerges, but because it falls largely outside the cone of influence, no robust interpretation can be drawn.

For the biological variables, the WPS for chl-a (Fig. 11d) shows a strong and statistically significant 1-year signal that persists across the full time series (1998–2024). FG_{chl} (Fig. 11e) exhibits a weaker but more continuous annual signal compared to Zone 1, although it weakens or disappears between 2002–2006 and again in 2018. As in the mesotrophic subregion, FP_{chl} (Fig. 11f) shows a weak but significant 1-year signal that persists through most of the time series, except for a short coherence gap between 2001 and 2003.

3.6. Quantile regression analyses

This section presents the main results of the QR analyses for SST and chl-a, and their associated frontal metrics across the two subregions: the AJ-WAG-CsCG system and the surrounding oligotrophic waters. Because trend strength and significance vary across spatial domains, the spatial frame of reference directly shapes the interpretation of long-term changes. Subregional analyses (Zones 1 and 2) may reveal more localised processes or attenuated trends compared with analyses at basin scales. This distinction is especially important when spatial heterogeneity is ecologically or physically meaningful, as it is in the Alborán Sea.

At the subregion influenced by the AJ-WAG-CsCG system (Zone 1), SST and their associated frontal metrics also show increasing trends along the study period (Fig. 12a–c, Table 1). However, the upward trend in the SST time series is statistically significant only at the 0.25th quartile (0.049 ± 0.018 °C km⁻¹ yr⁻¹), meanwhile the increasing trends at all three quartile for both frontal metrics (FG_{sst} and FP_{sst}) are statistically significant. For FG_{sst} , the slope increases with the quartile, showing a trend of 0.003 ± 0.001 °C km⁻¹ yr⁻¹ for the lowest temperature values, of 0.004 ± 0.002 °C km⁻¹ yr⁻¹ for the median trend analysis, and 0.008 ± 0.001 °C km⁻¹ yr⁻¹ for the highest temperature values (Table 1). For FP_{sst} , the strongest increase is observed at the 0.50th quartile (0.611 ± 0.087 % yr⁻¹), followed by the 0.25th

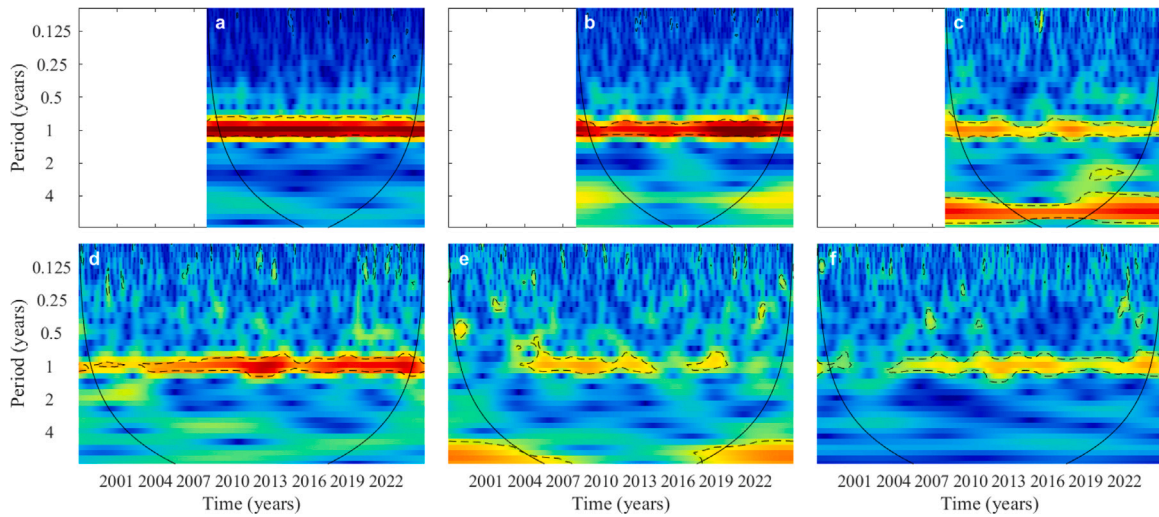


Fig. 10. Wavelet Power Spectrum (WPS) analysis for the mesotrophic subregion (Zone 1), showing: (a) Sea surface temperature (SST, °C) and its associated frontal metrics, (b) frontal gradient (FG_{sst} , °C km⁻¹) and (c) frontal probability (FP_{sst} , %), over the period 2008–2024 (top row); and (d) chlorophyll-*a* concentration (chl-*a*, mg m⁻³) and its associated frontal metrics, (e) frontal gradient (FG_{chl} , mg m⁻³ km⁻¹) and (f) frontal probability (FP_{chl} , %), over the 1998–2024 (bottom row). Colour shading indicates power intensity, from dark blue (low) to dark red (high). Black contours delineate the cone of influence. Dotted lines mark the $\alpha = 5\%$ significance level, estimated from 1000 bootstrapped Markov time series.

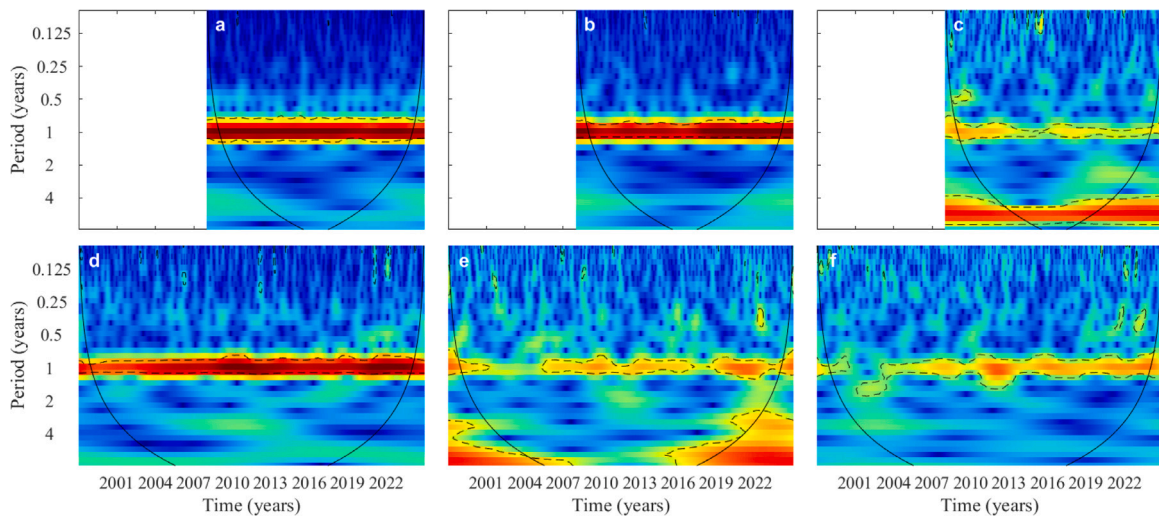


Fig. 11. Wavelet Power Spectrum (WPS) analysis for the oligotrophic subregion (Zone 2), showing: (a) Sea surface temperature (SST, °C) and its associated frontal metrics, (b) frontal gradient (FG_{sst} , °C km⁻¹) and (c) frontal probability (FP_{sst} , %), over the period 2008–2024 (top row); and (d) chlorophyll-*a* concentration (chl-*a*, mg m⁻³) and its associated frontal metrics, (e) frontal gradient (FG_{chl} , mg m⁻³ km⁻¹) and (f) frontal probability (FP_{chl} , %), over the period 1998–2024 (bottom row). Colour shading indicates power intensity, from dark blue (low) to dark red (high). Black contours delineate the cone of influence. Dotted lines mark the $\alpha = 5\%$ significance level, estimated from 1000 bootstrapped Markov time series.

($0.574 \pm 0.099 \text{ \% yr}^{-1}$) and 0.75th ($0.517 \pm 0.088 \text{ \% yr}^{-1}$) quartiles (Table 1). While values differ slightly, all trends fall within the same order of magnitude.

The subregion influenced by the AJ-WAG-CsCG system is the most productive area of the Alborán Sea, characterised by the highest chl-*a* concentrations (Fig. 12d–f, Table 1). Here, the chl-*a* time series shows the strongest decreasing trend observed across the entire region (Fig. 12d). The largest declines are found at the 0.25th and 0.50th percentiles, with trend values of $-0.006 \pm 9 \cdot 10^{-4}$ and $-0.007 \pm 0.002 \text{ mg m}^{-3} \text{ yr}^{-1}$, respectively; both are statistically significant. For FG_{chl} , the trend at the 0.25th percentile shows a slight increase, but it is not statistically significant. At the 0.50th and 0.75th percentiles, the trends are negative but of low magnitude and also not statistically significant (Fig. 12e; Table 1). FP_{chl} shows statistically significant decreasing trends at the 0.25th and 0.50th percentiles (Fig. 12f; Table 1). At the 0.25th percentile, the trend ($-0.061 \pm 0.020 \text{ \% yr}^{-1}$) is slightly lower

than that observed for the whole region (data not shown). For the median values, FP_{chl} decreases more slowly in this subregion, with a trend of $-0.053 \pm 0.019 \text{ \% yr}^{-1}$ (Table 1).

In the oligotrophic waters (Zone 2), SST and its associated frontal metrics show similar behaviour to that observed in the rest of the Alborán Sea (Fig. 13a–c). SST displays the highest increasing trend at the 0.50th quartile and the lowest at the 0.75th quartile (Fig. 13a). However, only the trend at the 0.25th quartile is statistically significant, with a value of $0.054 \pm 0.015 \text{ °C yr}^{-1}$ (Table 1). FG_{sst} shows increasing trend values (Fig. 13b), but, only the trend at the 0.75th quartile is statistically significant ($0.005 \pm 0.001 \text{ °C km}^{-1} \text{ yr}^{-1}$; Table 1). FP_{sst} shows a decreasing trend magnitude across quartiles, with statistically significant increases of $0.604 \pm 0.094 \text{ \% yr}^{-1}$ at the 0.25th quartile, $0.558 \pm 0.067 \text{ \% yr}^{-1}$ at the 0.50th quartile, and $0.551 \pm 0.072 \text{ \% yr}^{-1}$ at the 0.75th quartile (Fig. 13c, Table 1).

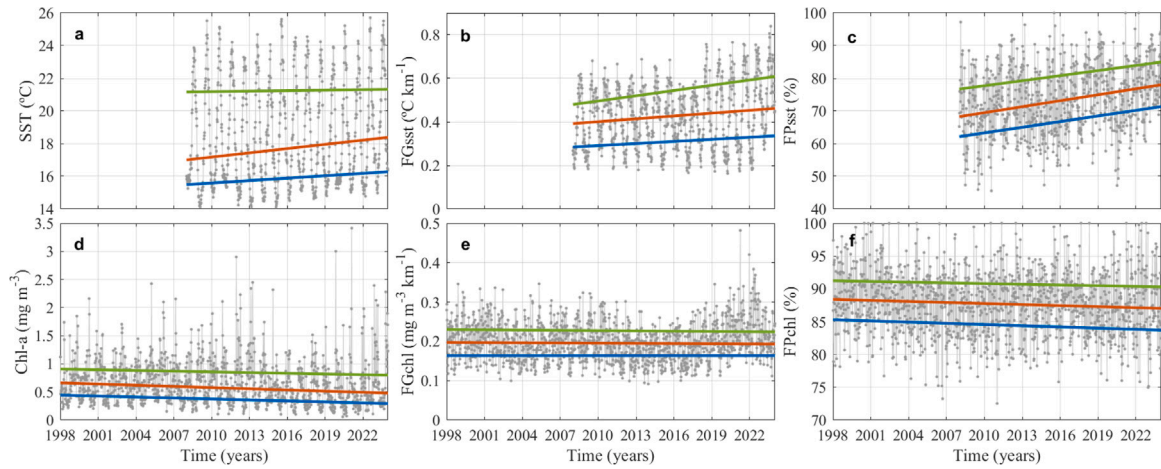


Fig. 12. Quantile regression analysis for the mesotrophic subregion (Zone 1), showing: (a) Sea surface temperature (SST, °C) and its associated frontal metrics, (b) frontal gradient (FG_{sst} , °C km⁻¹) and (c) frontal probability (FP_{sst} , %), over the period 2008–2024; and (d) chlorophyll-a concentration (chl-a, mg m⁻³) and its associated frontal metrics, (e) frontal gradient (FG_{chl} , mg m⁻³ km⁻¹) and (f) frontal probability (FP_{chl} , %), over the period 1998–2024. Grey dots represent weekly averages, and the grey line indicates the full time series. Trend lines correspond to the 0.25th (blue), 0.50th or median (red), and 0.75th (green) percentiles.

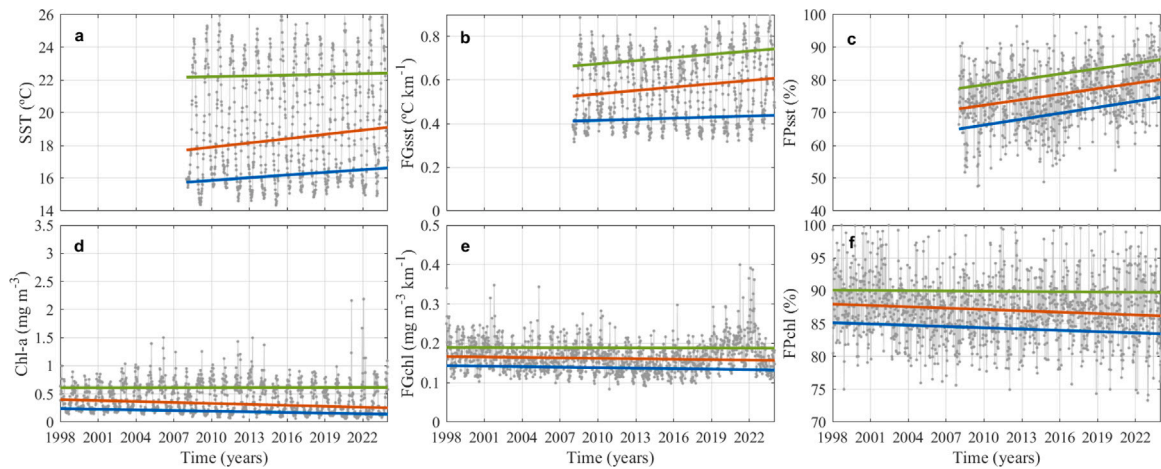


Fig. 13. Quantile regression analysis for the oligotrophic subregion (Zone 2), showing: (a) Sea surface temperature (SST, °C) and its associated frontal metrics, (b) frontal gradient (FG_{sst} , °C km⁻¹) and (c) frontal probability (FP_{sst} , %), over the period 2008–2024; and (d) chlorophyll-a concentration (chl-a, mg m⁻³) and its associated frontal metrics, (e) frontal gradient (FG_{chl} , mg m⁻³ km⁻¹) and (f) frontal probability (FP_{chl} , %), over the period 1998–2024. Grey dots represent weekly averages, and the grey line indicates the full time series. Trend lines correspond to the 0.25th (blue), 0.50th or median (red), and 0.75th (green) percentiles.

Here, the decreasing trends in chl-a concentration are similar to those observed for the entire region (data not shown), with statistically significant values only at the 0.25th and 0.50th percentiles ($-0.004 \pm 3 \cdot 10^{-4}$ and -0.006 ± 0.001 mg m⁻³ yr⁻¹, respectively). FG_{chl} trends remain of low magnitude (Fig. 13b), with the steepest decline, and the only statistically significant value, found at the 0.25th percentile ($-4.28 \cdot 10^{-4} \pm 1 \cdot 10^{-4}$ mg m⁻³ km⁻¹ yr⁻¹) (Table 1). As in the other subregions, FP_{chl} shows statistically significant decreasing trends only at the 0.25th and 0.50th percentiles, with values of -0.065 ± 0.024 and -0.069 ± 0.018 % yr⁻¹, respectively (Fig. 13f; Table 1).

4. Discussion

Global climate change projection scenarios identify the Mediterranean Sea as one of the regions most responsive to rising temperatures (e.g., Giorgi, 2006). This temperate region is expected to become warmer and drier (IPCC, 2021), and, being semi-enclosed, has limited exchange with the open ocean, making it particularly vulnerable to pollution and other human pressures, including overfishing (e.g., Hoegh-Guldberg et al., 2014). Although several remote sensing and

modelling studies have analysed climate change impacts on hydrodynamics and productivity in the Mediterranean Sea (e.g., Parras-Berrocal et al., 2020; El Hourany et al., 2021), none has specifically focused on the Alborán Sea, the most productive basin in the Mediterranean (Bosc et al., 2004), with the spatial and temporal resolution required to capture the frontal processes that characterise this region.

The Alborán Sea is a highly dynamic region where intense physical-biological coupling has long been documented (e.g., Sarhan et al., 2000; Macías et al., 2008; Navarro et al., 2011; Oguz et al., 2014). The circulation shaped by the AJ and the associated AJ-WAG-CsCG system sustains persistent frontal activity linked to upwelling and cross-frontal exchange (Bolado-Penagos et al., 2020), which inject nutrient-rich subsurface waters into the euphotic layer (Oguz et al., 2016; Sala et al., 2022). Understanding whether this coupling is being modified is therefore essential for anticipating future ecosystem functioning. Increased stratification, altered circulation, or changes in frontal behaviour could all influence the productivity of these features. Our results show that frontal dynamics and phytoplankton responses have evolved in parallel over recent decades, suggesting a shift in their historical coupling.

Table 1

Quantile regression slopes and 95% confidence intervals for the 0.25th, 0.50th (median), and 0.75th percentiles of sea surface temperature (SST, °C yr⁻¹), SST frontal gradient (FG_{sst} , °C km⁻¹ yr⁻¹), SST frontal probability (FP_{sst} , % yr⁻¹) (2008–2024), chlorophyll-*a* concentration (chl-*a*, mg m⁻³ yr⁻¹), chl-*a* frontal gradient (FG_{chl} , mg m⁻³ km⁻¹ yr⁻¹), and chl-*a* frontal probability (FP_{chl} , % yr⁻¹) (1998–2024) in the mesotrophic (Zone 1) and oligotrophic (Zone 2) subregions. Asterisks denote statistically significant trends ($p < 0.05$). C.I.: confidence interval.

Variable	Percentile	Zone 1: AJ-WAG-CsCG system		Zone 2: Oligotrophic waters	
		Slope	95% C.I.	Slope	95% C.I.
SST (°C yr ⁻¹)	0.25th	0.049* ± 0.018	[0.015, 0.083]	0.054* ± 0.015	[0.024, 0.088]
	0.50th	0.087 ± 0.042	[-0.004, 0.151]	0.086 ± 0.040	[0.000, 0.157]
	0.75th	0.010 ± 0.053	[-0.073, 0.120]	0.016 ± 0.048	[-0.061, 0.124]
FG_{sst} (°C km ⁻¹ yr ⁻¹)	0.25th	0.003* ± 0.001	[2 · 10 ⁻⁴ , 6 · 10 ⁻³]	0.002 ± 0.002	[-1 · 10 ⁻⁴ , 4 · 10 ⁻³]
	0.50th	0.004* ± 0.002	[6 · 10 ⁻⁴ , 8 · 10 ⁻³]	0.005 ± 0.003	[1 · 10 ⁻⁴ , 1 · 10 ⁻²]
	0.75th	0.008* ± 0.001	[0.004, 0.011]	0.005* ± 0.001	[0.002, 0.007]
FP_{sst} (% yr ⁻¹)	0.25th	0.574* ± 0.099	[0.378, 0.766]	0.604* ± 0.094	[0.418, 0.775]
	0.50th	0.611* ± 0.087	[0.425, 0.757]	0.558* ± 0.067	[0.436, 0.701]
	0.75th	0.517* ± 0.088	[0.327, 0.677]	0.551* ± 0.072	[0.408, 0.692]
Chl- <i>a</i> (mg m ⁻³ yr ⁻¹)	0.25th	-0.006* ± 9 · 10⁻⁴	[-0.007, -0.004]	-0.004* ± 3 · 10⁻⁴	[-0.005, -0.003]
	0.50th	-0.007* ± 0.002	[-0.011, -0.003]	-0.006* ± 0.001	[-0.008, -0.003]
	0.75th	-0.004 ± 0.002	[-0.009, 0.001]	1.96 · 10 ⁻⁴ ± 0.001	[-0.002, 0.003]
FG_{chl} (mg m ⁻³ km ⁻¹ yr ⁻¹)	0.25th	4.20 · 10 ⁻⁶ ± 2 · 10 ⁻⁴	[-3 · 10 ⁻⁴ , 4 · 10 ⁻⁴]	-4.28 · 10⁻⁴* ± 1 · 10⁻⁴	[-7 · 10 ⁻⁴ , -2 · 10 ⁻⁴]
	0.50th	-1.52 · 10 ⁻⁴ ± 2 · 10 ⁻⁴	[-6 · 10 ⁻⁴ , 3 · 10 ⁻⁴]	-3.72 · 10 ⁻⁴ ± 1 · 10 ⁻⁴	[-6 · 10 ⁻⁴ , -1 · 10 ⁻⁴]
	0.75th	-2.30 · 10 ⁻⁴ ± 1 · 10 ⁻⁴	[-7 · 10 ⁻⁴ , 3 · 10 ⁻⁴]	-6.20 · 10 ⁻⁵ ± 2 · 10 ⁻⁴	[-4 · 10 ⁻⁴ , 3 · 10 ⁻⁴]
FP_{chl} (% yr ⁻¹)	0.25th	-0.061* ± 0.020	[-0.103, -0.022]	-0.065* ± 0.024	[-0.114, -0.026]
	0.50th	-0.053* ± 0.019	[-0.090, -0.019]	-0.069* ± 0.018	[-0.104, -0.032]
	0.75th	-0.036 ± 0.025	[-0.087, 0.014]	-0.014 ± 0.023	[-0.068, 0.023]

4.1. Frontal processes in the Alborán Sea

To study the frontal processes that characterise the Alborán Sea, we applied the CED algorithm to characterise the seasonal behaviour and spatial structure of surface fronts, focusing on mesoscale frontal variability. It is important to note that frontal probability (FP) and frontal gradient (FG) provide different and complementary information: FP indicates how frequently a pixel is classified as a front, whereas FG describes the intensity of the gradient between the frontal pixel and its surroundings. As such, high FP values do not necessarily imply strong physical or biological fronts if FG remains low. In line with expectations, we found that all four frontal metrics (FG_{sst} , FP_{sst} , FG_{chl} , and FP_{chl}) display a similar seasonal cycle across the Alborán Sea (Figs. 3–6). Values increase progressively from winter, reaching a maximum between late spring and early summer (June–July), followed by a gradual decline towards autumn and reaching minimum levels in winter. Moreover, this analysis clearly divides the Alborán Sea into two regions with different dynamics (Fig. 1) that should be studied independently: the mesotrophic waters influenced by the AJ-WAG-CsCG system, and the surrounding oligotrophic waters.

In the mesotrophic waters, influenced by the AJ-WAG-CsCG system (Zone 1), frontal activity remains elevated for most of the year and throughout the study period (Figs. 3–8). Following the variability pattern of SST (Fig. 7a), FG_{sst} increases gradually from winter to summer (Figs. 3 and 7b), reflecting increasing thermal contrasts across the region. While upwelling processes associated with frontal activity and the CsCG continue to bring relatively cold waters year-round, the surrounding stratified waters progressively warm due to seasonal heating, enhancing the horizontal temperature gradient. In fact, FP_{sst} remains high throughout the year, indicating persistent frontal activity even when the temperature gradient is less pronounced (Figs. 4 and 7c).

FG_{chl} increases markedly during the bloom period (Fig. 5), but shows much greater interannual variability, especially in winter and early spring (Fig. 7d, e). In fact, the WPS analysis reveals a strong seasonal signal only during 2005–2013 and 2016–2018 (Fig. 10e). This variability likely reflects the complex interplay between nutrient availability, light limitation, and top-down control across seasons. The hydrodynamic setting of this region suggests a pulsating and relatively continuous supply of nutrients into the photic layer, especially through frontal upwelling and coastal cyclonic activity (Sala et al., 2022).

However, phytoplankton growth generally remains partially limited by light availability during winter, which may delay the onset of the bloom despite the early presence of nutrients. In this context, the increase in irradiance at the end of winter becomes a key driver of bloom initiation (e.g., Hopkins et al., 2021). On the other hand, the presence of physical nutrient inputs does not imply the absence of nutrient limitation. Nitrogen limitation is known to occur in the Alborán Sea (Ramírez et al., 2005; Belgacem et al., 2021), and after the bloom period, a decline in biomass would be expected due to nutrient depletion and strong zooplankton grazing pressure (e.g., Irigoien et al., 2005). In fact, sustained high irradiance in the uppermost layers, even in the presence of nutrients, is unlikely to support further growth due to increased physiological stress in some phytoplankton groups. Moreover, following the rapid growth phase and intense grazing during spring, the community may require time to recover. As a result, even if nutrient inputs persist through frontal activity, biomass levels do not return to their spring peak, suggesting that summer conditions reflect not only resource limitation, but also the aftermath of ecological pressure. Although FP_{chl} remains consistently high throughout the year, these values are ecologically relevant, as FG_{chl} also remains relatively high (minimum values around 0.20 mg m⁻³ km⁻¹), indicating persistent and structured horizontal gradients in surface chlorophyll.

In the oligotrophic waters (Zone 2), following the same variability pattern of SST (Fig. 8a), FG_{sst} and FP_{sst} increase from winter to summer but both remain weaker and more spatially diffuse compared to the values observed in the mesotrophic region (Figs. 3 and 4). When analysing the interannual variability of FG_{sst} , although variability is lower, the range of values (~0.35–0.85 °C km⁻¹; red points in Fig. 8b) is higher than those observed in Zone 1 (0.20–0.80 °C km⁻¹; red points in Fig. 7b). This apparent inconsistency seems to be an artefact of how the averages are computed over fixed spatial masks, which do not account for the fact that the physical structures studied here are dynamic and shift in space with the seasonal cycle. Moreover, it is worth remembering that the spatial delimitation of both zones is based on the 27-year mean chl-*a* distribution, rather than SST. As a result, the mismatch between Zone 1 and Zone 2 likely reflects both factors, the fixed mask and the variable SST structure. Furthermore, the maximum FG_{sst} values observed in summer may be influenced by the strong SST maxima that characterise the WAG (see Fig. A3).

FG_{chl} in Zone 2 shows a weak seasonal cycle, increasing during spring and early summer, particularly near coastal areas, but rarely

exceeding $0.20 \text{ mg m}^{-3} \text{ km}^{-1}$ (Figs. 5 and 11e). Moreover, it shows limited interannual variability, although in the months close to the upwelling period this variability is greater, reaching values of $\sim 0.30 \text{ mg m}^{-3} \text{ km}^{-1}$ (Fig. 8e). FP_{chl} follows the same seasonal pattern but remains relatively low across most of the region and lacks the spatial coherence observed in Zone 1 (Fig. 6). Interannually, FP_{chl} shows lower variability than in Zone 1, but remains within the same range (80%–95%) and also reaches a maximum during summer (Fig. 8f). These patterns indicate that chl-*a* gradients are generally weak and poorly structured across the region, and that even when frontal pixels are detected, the underlying gradient strength is minimal (limiting the ecological significance of FP_{chl} in this zone). Sala et al. (2022), analysing the trajectory of the AJ following a Lagrangian drifter, observed that as the AJ flows through the Alborán Sea, it progressively subducts, and the input of nutrient-rich colder waters associated with frontal processes does not reach the most superficial layers. This likely contributes to the decreasing presence of frontal pixels (both SST and chl-*a*) on the eastern side of the basin, although it does not rule out the existence of frontal activity at depth, beyond the reach of satellite detection. In this region, wind forcing is probably the main driver of frontal pixel detection at the surface.

4.2. Bloom phenological shifts and interannual trends

The seasonal evolution of SST and chl-*a* (Fig. 9) reveals a consistent temporal mismatch between physical and biological processes across both regions. This desynchronisation, together with the opposing trends in both variables and their frontal metrics (i.e., temperature fronts are getting stronger, while blooms are weakening), points out to an increasing decoupling between physical forcing and phytoplankton response over time. Surface-based SST fronts may become sharper due to increased thermal gradients, but this often results from stronger stratification, which in turn reduces vertical nutrient supply to the photic zone (Behrenfeld et al., 2006; Doney, 2006). Therefore, although physical frontal activity may persist or even intensify, its capacity to sustain biological production is diminished (Lévy et al., 2012; Mahadevan, 2016).

As previously discussed, phytoplankton dynamics in the mesotrophic subregion (Zone 1) are far from uniform. Although SST follows a clear seasonal cycle and reaches its annual minimum in late winter (Figs. 7a and 9a), the timing, duration, and intensity of phytoplankton blooms vary markedly from year to year (Figs. 7d and 9c). In temperate regions, blooms are typically expected in early spring (March–May), when light availability increases (Estrada, 1996; Agustí and Duarte, 2000). However, in our study region, peaks can occur as early as January or February, while in other years the increase is broader or delayed, extending into late spring (Fig. 9a, c). This variability is also captured in FG_{chl} , which increases markedly during the bloom period (Fig. 5) but shows much higher interannual dispersion in winter and early spring (Fig. 7d, e). In this study, bloom phenology reflects a complex and dynamic interplay between physical forcing, nutrient availability, light limitation, and top-down control; phytoplankton dynamics here are highly variable. Ultimately, the bloom likely results from a combination of sustained nutrient availability and moderate stratification, which support phytoplankton accumulation before grazing and nutrient depletion take over.

Although it is not easy to detect, some phenological shifts are taking place in the subregion influenced by the AJ–WAG–CsCG system. There is a gradual warming trend throughout the later years of the time series, with temperatures appearing slightly higher across all seasons (Fig. 9a). The QR analysis confirms that SST in this region is increasing, although only the increase in the lowest temperatures is statistically significant, with values of $0.049 \pm 0.018 \text{ }^{\circ}\text{C yr}^{-1}$ (Fig. 10a, Table 1). However, both FG_{sst} and FP_{sst} show statistically significant increasing trends, indicating that not only are temperature contrasts intensifying across

the region, but also those frontal structures are becoming more frequent and persistent over time (Fig. 10b–c, Table 1).

Meanwhile, a decrease in chl-*a* concentration, reaching values below 0.20 mg m^{-3} around summer, was also observed (Fig. 9c). The trend analysis confirms this pattern, with statistically significant decreasing trends for the lowest and median values (0.25th and 0.50th percentiles), of $-0.006 \pm 9 \cdot 10^{-4}$ and $-0.007 \pm 0.002 \text{ mg m}^{-3} \text{ yr}^{-1}$, respectively (Table 1). Concurrently, FG_{chl} and FP_{chl} also show decreasing trends over the study period, although only FP_{chl} at the 0.25th and 0.50th percentiles are statistically significant. The negative trend at the 0.25th percentile likely reflects a progressive intensification of summer oligotrophy, consistent with the observed biomass minimum during this season. In contrast, the decreasing trend in the median values points to a broader decline across the annual cycle, potentially affecting transitional periods such as early spring or late summer, where blooms are weaker or more variable. Taken together, these results indicate that the mesotrophic subregion of the Alborán Sea is experiencing long-term biological weakening, even in the presence of increasing SST frontal activity.

Similar phenological changes were observed in the oligotrophic subregion (Zone 2). The QR analysis confirms the observed gradual warming over the latter years (Fig. 9b), with increasing trends for SST and its frontal metrics (Fig. 11a–c and Table 1). As in Zone 1, the increasing SST trend is only statistically significant for the 0.25th percentile, indicating that SST is increasing at a rate of $0.054 \pm 0.015 \text{ }^{\circ}\text{C yr}^{-1}$ during colder periods of the year (i.e., winter). This warming trend rate is slightly higher than in the frontal region, likely because Zone 2 is dynamically more stable and subject to weaker vertical mixing. The reduced exchange with deeper, colder waters may allow the surface layer to accumulate heat more efficiently, potentially strengthening near-surface stratification over time (Behrenfeld et al., 2006; Somavilla et al., 2017). FG_{sst} only showed a significant trend for the 0.75th percentile ($0.005 \pm 0.001 \text{ }^{\circ}\text{C km}^{-1} \text{ yr}^{-1}$), indicating that the most intense thermal gradients are becoming stronger over time (Fig. 11b, Table 1). However, all FP_{sst} percentiles show a significant increasing trend (Table 1); but we must not forget that the pixels identified as frontal in this region may not have strong dynamical implications, because the FG_{sst} values they are associated with are relatively low. This may indicate that although frontal pixels are detected more frequently, many of them correspond to weak thermal contrasts, and may reflect surface heating patterns rather than physically active frontal zones.

In Zone 2, the biological metrics showed similar behaviour to the one observed in Zone 1. Chl-*a* concentration exhibited a statistically significant decreasing trend at the 0.25th and 0.50th percentiles (Fig. 11d, Table 1), mirroring the decline observed in the mesotrophic region. However, the rates of change were lower, with values of $-0.004 \pm 3.10 \cdot 10^{-4}$ and $-0.006 \pm 0.001 \text{ mg m}^{-3} \text{ yr}^{-1}$, respectively. FG_{chl} declined slowly over time, with a statistically significant trend at the lower quantile ($-4.28 \cdot 10^{-4} \pm 1.10 \cdot 10^{-4} \text{ mg m}^{-3} \text{ km}^{-1} \text{ yr}^{-1}$). In addition, FP_{chl} also declined over time, with statistically significant trends at the lower and median quantiles (-0.065 ± 0.024 and $-0.069 \pm 0.018 \text{ } \%$ yr^{-1} , respectively). These results indicate that, regardless of the local physical setting, baseline and median chl-*a* values are decreasing across the oligotrophic subregion.

Despite the contrasting physical regimes of the two subregions, the biological response appears remarkably consistent. This convergence suggests that, although the drivers may differ locally (i.e., mismatches between bloom timing and frontal development in the frontal zone and nutrient stress linked to stratification in the east), the underlying pattern reflects a basin-wide weakening of the physical–biological coupling. The fact that this decline occurs even in areas where frontal activity is intensifying (Zone 1) reinforces the view that the historical linkage between physical frontal structures and phytoplankton response, evidenced by consistent declines in chl-*a* and in the spatial coherence of biologically derived fronts (FP_{chl}), is weakening throughout the Alborán Sea. Similar decoupling between physical drivers and

biological responses has already been documented in other dynamic marine systems, including the California Current (Jamil et al., 2023) and the Southern Ocean (Chapman et al., 2020). Such long-term biological weakening has potential implications for the wider ecosystem. A reduction in phytoplankton biomass may lower food availability for zooplankton and other primary consumers, potentially triggering cascading effects through the pelagic food web. In the Alborán Sea, where productivity hotspots support small pelagic fish and higher trophic levels, a sustained decline in phytoplankton could ultimately influence recruitment, trophic transfer efficiency and regional fisheries productivity (Palomera et al., 2007).

A recent study by El Hourany et al. (2021), focused on the entire Mediterranean Sea, reported a significant increasing trend in SST of 0.036 ± 0.008 °C yr⁻¹, and a non-significant decreasing trend in chl-*a* of -0.0005 ± 0.0014 mg m⁻³ yr⁻¹. Their SST and chl-*a* trends are approximately 2.50 and 12 times smaller, respectively, than those found in this study for the Alborán Sea (0.090 ± 0.038 °C yr⁻¹ and -0.006 ± 0.002 mg m⁻³ yr⁻¹, 0.50th percentile, which is roughly comparable to a linear estimate used in their analysis). El Hourany et al. (2021) based their analysis on an 18-year satellite dataset of SST and chl-*a* at 4-km resolution, using monthly averages. In addition, their regionalisation, derived from a SOM-based classification, treats the Alborán Sea as a single homogeneous unit. These differences highlight the importance of regional-scale analyses and underscore the critical role of spatial and temporal resolution in detecting localised trends in oceanographic variables. Temporal and spatial resolution is not a technical detail, but a key analytical decision that can determine whether emerging ecological shifts are detected or missed (e.g., Henson et al., 2018; Caballero et al., 2020; Yang and Fischer, 2025).

Recent observations by Vargas-Yáñez et al. (2022), based on multiple time series of mixed layer depth (MLD) and CTD profiles across the Spanish Mediterranean, reported no significant trends in MLD over the past two decades. This would suggest an absence of increased stratification at the basin scale. Yet, the lack of change in this bulk metric does not rule out shifts in upper-layer dynamics at finer spatial or temporal scales, as suggested by the patterns described in this study. In highly dynamic regions like the Alborán Sea, differential warming between inflowing Atlantic waters and resident Mediterranean waters may still strengthen near-surface density gradients. Such changes might escape detection through traditional MLD estimates, yet still reduce vertical mixing and nutrient supply to the surface, contributing to the observed weakening in phytoplankton responses despite apparently stable stratification. However, confirming whether near-surface stratification has intensified would require subsurface observations, as satellite-derived SST gradients alone cannot capture the full vertical structure of the water column; the mechanism remains plausible but cannot be demonstrated without *in situ* data.

Nonetheless, satellite observations alone cannot resolve the subsurface processes that regulate nutrient supply and vertical exchange, meaning that some frontal features, particularly those that intensify at depth, may be only weakly expressed at the surface. A clearer understanding of frontal variability and the emerging biological decoupling would therefore benefit from complementary observations. Recent studies integrating satellite data with glider sections, biogeochemical sampling and high-resolution modelling have shown that this multiplatform perspective improves the characterisation of vertical motions and tracer pathways across frontal systems (Cotroneo et al., 2016; Mena et al., 2016; Tzortzis et al., 2021, 2023). In the Alborán Sea, where submesoscale dynamics are important yet difficult to constrain from space, such approaches could help determine how changes in stratification, circulation or the Atlantic Jet influence frontal activity. Moreover, high-resolution physical–ecological models are especially valuable for testing different forcing scenarios over multi-decadal timescales and, when combined with targeted *in situ* measurements, could help clarify the mechanisms behind the physical–biological mismatch identified in this study.

5. Conclusions

This study provides the first high-resolution, long-term assessment of frontal dynamics and associated phytoplankton responses in the Alborán Sea. By applying the Canny Edge Detector algorithm to 17 years of SST and 27 years of chl-*a* satellite data, we characterised both the seasonal patterns and interannual trends of physical–biological coupling in this complex transitional system. Our results show that while SST frontal structures are strengthening and becoming more frequent, phytoplankton blooms are weakening and undergoing shifts in timing. This decoupling suggests that historical links between surface fronts and productivity are eroding, possibly due to enhanced stratification and reduced nutrient supply. Notably, this trend is observed across both mesotrophic and oligotrophic subregions, pointing to a basin-wide decline in biological productivity.

Satellite data, even at kilometre-scale resolution, cannot resolve subsurface vertical motions or the three-dimensional components of submesoscale (<10 km) frontal processes that mediate nutrient upwelling and strongly influence phytoplankton growth. These unresolved dynamics likely contribute to the physical–biological mismatch detected here, underscoring the need for fine-scale, regionally resolved analyses to detect early signs of ecological change. Future work integrating subsurface observations (e.g., glider transects) together with high-resolution physical–ecological models such as ROMS or MITgcm will be essential to disentangle the relative roles of stratification, circulation changes and nutrient limitation in driving this decoupling. Such knowledge is critical to assess biodiversity changes across trophic levels, from primary producers to higher consumers, and consequently to determine their potential impacts on fisheries.

CRedit authorship contribution statement

Iria Sala: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Isabel Caballero:** Writing – review & editing. **Marina Bolado-Penagos:** Writing – review & editing. **Gabriel Navarro:** Writing – review & editing, Methodology, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Consensus for an initial deep literature review and ChatGPT (OpenAI, GPT-4) to assist with language editing and refinement. After using these tools, the authors carefully reviewed, verified, and edited all generated content and take full responsibility for the final version of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

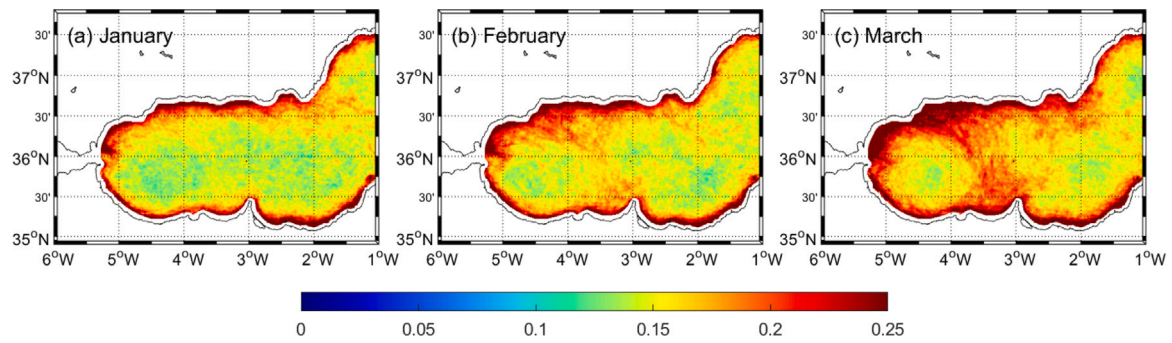


Fig. A1. Winter monthly climatology of chlorophyll-*a* frontal gradient (FG_{chl} , $\text{mg m}^{-3} \text{ km}^{-1}$) for the 1998–2024 period.

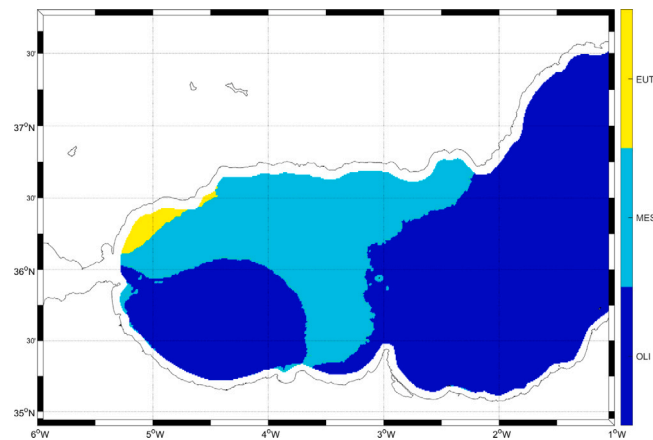


Fig. A2. Spatial organisation of the study region based on chlorophyll-*a* (chl-*a*) concentration trophic thresholds: OLI, $\text{chl-}a < 0.50 \text{ mg m}^{-3}$; MES, $0.50 \leq \text{chl-}a < 1.00 \text{ mg m}^{-3}$; EUT, $\text{chl-}a \geq 1.00 \text{ mg m}^{-3}$.

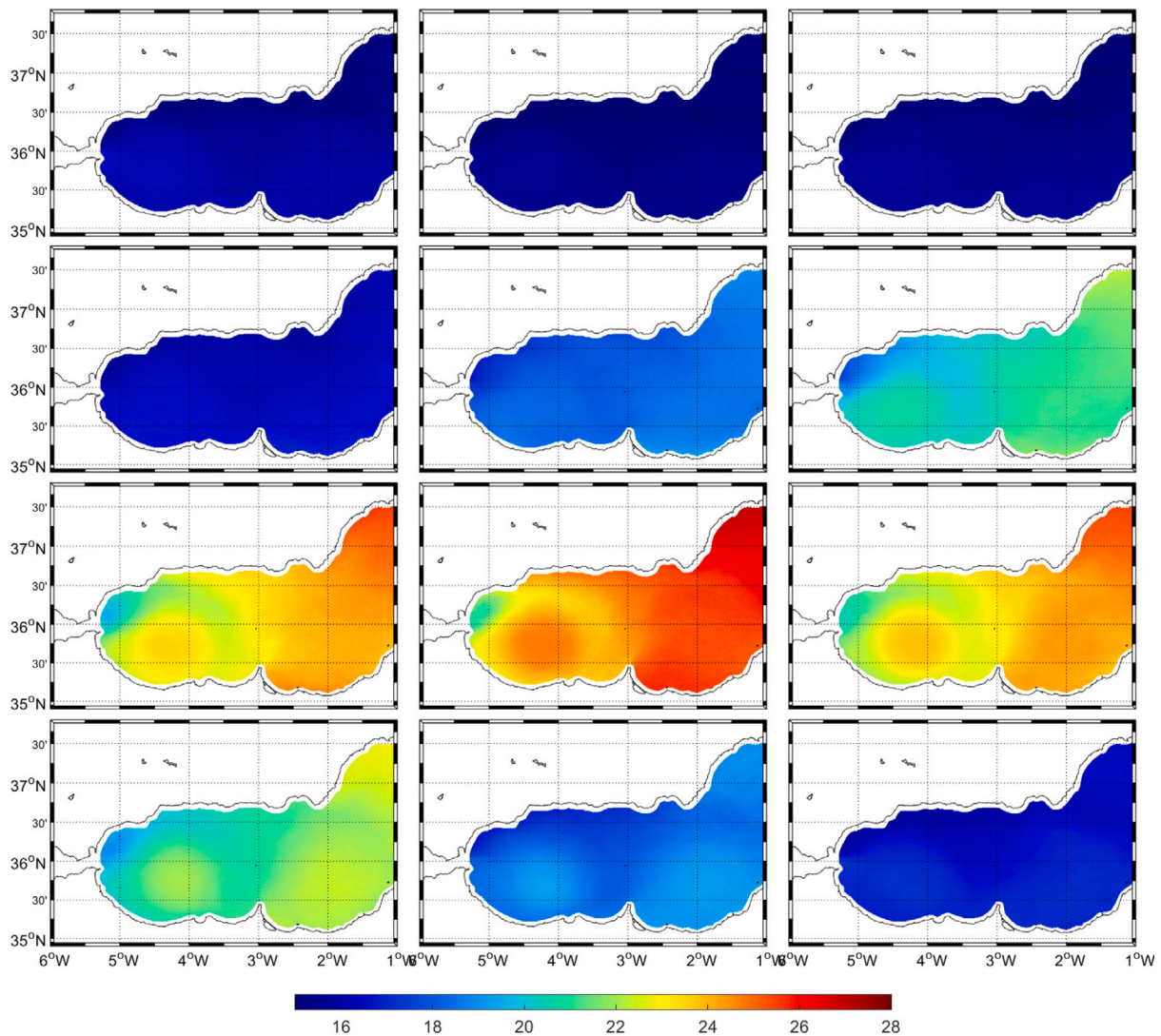


Fig. A3. Monthly climatology of sea surface temperature (SST, °C) for the 2008–2024 period.

Data availability

All datasets used in this study are publicly available through open-access repositories (e.g., Copernicus Marine Service, NASA OceanColor).

References

- Abdellaoui, B., Falcini, F., Baibai, T., Karim Hilmi, K., Ettahiri, O., Santoleri, R., Houssa, R., Nhala, H., Er-Raioui, H., Oukhattar, L., 2024. Spatial pattern of sea surface temperature and chlorophyll-a trends in relation to hydrodynamic processes in the Alborán Sea. *Mediterr. Mar. Sci.* 25, 136–150. <http://dx.doi.org/10.12681/mms.30268>.
- Agustí, S., Duarte, C.M., 2000. Strong seasonality in phytoplankton cell lysis in the NW Mediterranean littoral. *Limnol. Oceanogr.* 45, 940–947. <http://dx.doi.org/10.4319/lo.2000.45.4.0940>.
- Allen, J.T., Smeed, D.A., Tintoré, J., Ruiz, S., 2001. Mesoscale subduction at the Almería-Oran front: Part 1: Ageostrophic flow. *J. Mar. Syst.* 30, 263–285. [http://dx.doi.org/10.1016/S0924-7963\(01\)00062-8](http://dx.doi.org/10.1016/S0924-7963(01)00062-8).
- Arnone, R.A., Nero, R.W., Jech, J.M., De Palma, I., 1990. Acoustic imaging of biological and physical processes within Gulf Stream meanders. *Eos, Trans. Am. Geophys. Union* 71, <http://dx.doi.org/10.1029/EO071i029p00982-01>, 982–982.
- Behrenfeld, M.J., O'Malley, R.T., Siegel, D.A., McClain, C.R., Sarmiento, J.L., Feldman, G.C., Milligan, A.J., Falkowski, P.G., Letelier, R.M., Boss, E.S., 2006. Climate-driven trends in contemporary ocean productivity. *Nature* 444, 752–755. <http://dx.doi.org/10.1038/nature05317>.
- Belgacem, M., Schroeder, K., Barth, A., Troupin, C., Pavoni, B., Raimbault, P., Garcia, N., Borghini, M., Chiggiato, J., 2021. Climatological distribution of dissolved inorganic nutrients in the western Mediterranean Sea (1981–2017). *Earth Syst. Sci. Data* 13, 5915–5949. <http://dx.doi.org/10.5194/essd-13-5915-2021>.
- Belkin, I.M., 2002. Front. In: Nybakken, J.W., Broenkow, W.W., Vallier, T.L. (Eds.), *Interdisciplinary Encyclopedia of Marine Sciences*. Grolier, Danbury, CT, pp. 433–436.
- Belkin, I.M., Cornillon, P.C., Sherman, K., 2009. Fronts in large marine ecosystems. *Prog. Oceanogr.* 81, <http://dx.doi.org/10.1016/j.pocean.2009.04.015>.
- Belkin, I.M., O'Reilly, J.E., 2009. An algorithm for oceanic front detection in chlorophyll and SST satellite imagery. *J. Mar. Syst.* 78, <http://dx.doi.org/10.1016/j.jmarsys.2008.11.018>.
- Bolado-Penagos, M., González, C.J., Chioua, J., Sala, I., Jesús Gomiz-Pascual, J., Vázquez, Á., Bruno, M., 2020. Submesoscale processes in the coastal margins of the Strait of Gibraltar. The Trafalgar - Alboran connection. *Prog. Oceanogr.* 181, 102219. <http://dx.doi.org/10.1016/j.pocean.2019.102219>.
- Bolado-Penagos, M., Sala, I., Gomiz-Pascual, J.J., Romero-Cózar, J., González-Fernández, D., Reyes-Pérez, J., Vázquez, Á., Bruno, M., 2021. Revisiting the effects of local and remote atmospheric forcing on the Atlantic Jet and Western Alboran Gyre dynamics. *J. Geophys. Res.: Ocean.* 126, e2020JC016173. <http://dx.doi.org/10.1029/2020JC016173>.

- Bosc, E., Bricaud, A., Antoine, D., 2004. Seasonal and interannual variability in algal biomass and primary production in the Mediterranean Sea, as derived from 4 years of SeaWiFS observations. *Glob. Biogeochem. Cycles* 18, <http://dx.doi.org/10.1029/2003GB002034>.
- Buongiorno Nardelli, B., Tronconi, C., Pisano, A., Santoleri, R., 2013. High and ultra-high resolution processing of satellite sea surface temperature data over southern European seas in the framework of MyOcean project. *Remote Sens. Environ.* 129, 1–16. <http://dx.doi.org/10.1016/j.rse.2012.10.012>.
- Caballero, I., Fernández, R., Escalante, O.M., Mamán, L., Navarro, G., 2020. New capabilities of Sentinel-2A/B satellites combined with in situ data for monitoring small harmful algal blooms in complex coastal waters. *Sci. Rep.* 10, 8743. <http://dx.doi.org/10.1038/s41598-020-65600-1>.
- Campbell, J.W., 1995. The lognormal distribution as a model for bio-optical variability in the sea. *J. Geophys. Res.: Ocean.* 100, 13237–13254. <http://dx.doi.org/10.1029/95JC00458>.
- Canny, J., 1986. A computational approach to edge detection. *IEEE Trans. Pattern Anal. Mach. Intell. PAMI-8*, 679–698. <http://dx.doi.org/10.1109/TPAMI.1986.4767851>.
- Capet, X., McWilliams, J.C., Molemaker, M.J., Shchepetkin, A.F., 2008. Mesoscale to submesoscale transition in the California current system. part II: Frontal processes. *J. Phys. Oceanogr.* 38, 44–64. <http://dx.doi.org/10.1175/2007JPO3672.1>.
- Castelao, R.M., Barth, J.A., Mavor, T.P., 2005. Flow-topography interactions in the northern California current system observed from geostationary satellite data. *Geophys. Res. Lett.* 32, <http://dx.doi.org/10.1029/2005GL024401>.
- Cazelles, B., Chavez, M., Berteaux, D., Ménard, F., Vik, J.O., Jenouvrier, S., Stenseth, N.C., 2008. Wavelet analysis of ecological time series. *Oecologia* 156, 287–304. <http://dx.doi.org/10.1007/s00442-008-0993-2>.
- Cazelles, B., Stone, L., 2003. Detection of imperfect population synchrony in an uncertain world. *J. Anim. Ecol.* 72, 953–968. <http://dx.doi.org/10.1046/j.1365-2656.2003.00763.x>.
- Chapman, C.C., Lea, M.A., Meyer, A., Sallée, J.B., Hindell, M., 2020. Defining Southern Ocean fronts and their influence on biological and physical processes in a changing climate. *Nat. Clim. Chang.* 10, 209–219. <http://dx.doi.org/10.1038/s41558-020-0705-4>.
- Cotroneo, Y., Aulicino, G., Ruiz, S., Pascual, A., Budillon, G., Fusco, G., Tintoré, J., 2016. Glider and satellite high resolution monitoring of a mesoscale eddy in the algerian basin: Effects on the mixed layer depth and biochemistry. *J. Mar. Syst.* 162, 73–88. <http://dx.doi.org/10.1016/j.jmarsys.2015.12.004>.
- Crepon, M., 1965. Influence de la pression atmosphérique sur le niveau moyen de la méditerranée occidentale et sur le flux à travers le détroit de gibraltar. *Cah. Océanographiques* 1, 15–32.
- Doney, S.C., 2006. Plankton in a warmer world. *Nature* 444, 695–696. <http://dx.doi.org/10.1038/444695a>.
- El Hourany, R., Mejia, C., Faour, G., Crépon, M., Thiria, S., 2021. Evidencing the impact of climate change on the phytoplankton community of the Mediterranean Sea through a bioregionalization approach. *J. Geophys. Res.: Ocean.* 126, e2020JC016808. <http://dx.doi.org/10.1029/2020JC016808>.
- Estrada, M., 1996. Primary production in the northwestern Mediterranean. *Sci. Mar.* 60, 55–64.
- Falcini, F., Palatella, L., Cuttitta, A., Nardelli, B.B., Lacorata, G., Lanotte, A.S., Patti, B., Santoleri, R., 2015. The role of hydrodynamic processes on anchovy eggs and larvae distribution in the Sicily Channel (Mediterranean Sea): A case study for the 2004 data set. *PLOS ONE* 10, e0123213. <http://dx.doi.org/10.1371/journal.pone.0123213>.
- Garcia-Jove, M., Murre, B., Zarokanellos, N.D., Lermusiaux, P.F.J., Rudnick, D.L., Tintoré, J., 2022. Frontal dynamics in the Alboran Sea: 2. Processes for vertical velocities development. *J. Geophys. Res.: Ocean.* 127, <http://dx.doi.org/10.1029/2021JC017428>.
- Giorgi, F., 2006. Climate change hot-spots. *Geophys. Res. Lett.* 33, L08707. <http://dx.doi.org/10.1029/2006GL025734>.
- Gregg, M.C., 1987. Diapycnal mixing in the thermocline: A review. *J. Geophys. Res.* 92, 5249–5286. <http://dx.doi.org/10.1029/JC092iC05p05249>.
- Grinsted, A., Moore, J.C., Jevrejeva, S., 2004. Application of the cross wavelet transform and wavelet coherence to geophysical time series. *Nonlinear Process. Geophys.* 11, 561–566. <http://dx.doi.org/10.5194/npg-11-561-2004>.
- Henson, S.A., Cole, H.S., Hopkins, J., Martin, A.P., Yool, A., 2018. Detection of climate change-driven trends in phytoplankton phenology. *Global Change Biol.* 24, <http://dx.doi.org/10.1111/gcb.13886>.
- Hoegh-Guldberg, O., Cai, R., Poloczanska, E., Brewer, P., Sundby, Hilmi, K., Fabry, V., Jung, S., 2014. The ocean. In: Barros, V.R., Field, C.B., Dokken, D.J., Mastrandrea, M.D., Mach, K.J., Bilir, T.E., Chatterjee, M., Ebi, K.L., Estrada, Y.O., Genova, R.C., Girma, B., Kissel, E.S., Levy, A.N., Maccracken, S., Mastrandrea, P.R., White, L.L. (Eds.), *Climate Change 2014: Impacts, Adaptation, and Vulnerability. Part B: Regional Aspects. Contribution of Working Group II To the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, pp. 1655–1731.
- Hopkins, J.E., Palmer, M.R., Poulton, A.J., Hickman, A.E., Sharples, J., 2021. Control of a phytoplankton bloom by wind-driven vertical mixing and light availability. *Limnol. Oceanogr.* 66, 1926–1949. <http://dx.doi.org/10.1002/lno.11734>.
- Horner-Devine, A.R., Hetland, R.D., MacDonald, D.G., 2015. Mixing and transport in coastal river plumes. *Annu. Rev. Fluid Mech.* 47, 569–594. <http://dx.doi.org/10.1146/annurev-fluid-010313-141408>.
- Ignatiades, L., 2005. Scaling the trophic status of the Aegean Sea, eastern Mediterranean. *J. Sea Res.* 54, 51–57. <http://dx.doi.org/10.1016/j.seares.2005.02.010>.
- IPCC, 2021. Summary for policymakers. In: Masson-Delmotte, V., Zhai, P., Pirani, A., Connors, S.L., Péan, C., Berger, S., Caud, N., Chen, Y., Goldfarb, L., Gomis, M.I., Huang, M., Leitzell, K., Lonnoy, E., Matthews, J.B.R., Maycock, T.K., Waterfield, T., Yelekçi, O., Yu, R., Zhou, B. (Eds.), *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I To the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press.
- Irigoin, X., Flynn, K.J., Harris, R.P., 2005. Phytoplankton blooms: A 'loophole' in microzooplankton grazing impact? *J. Plankton Res.* 27, 313–321. <http://dx.doi.org/10.1093/plankt/fbi011>.
- Jamil, A.L., Kavanaugh, M.T., Spitz, Y.H., 2023. Drivers of physical and biological frontal variability in the northern California current system. *J. Geophys. Res.: Ocean.* 128, e2022JC019408. <http://dx.doi.org/10.1029/2022JC019408>.
- La Violette, P.E., Tintoré, J., Font, J., 1990. The surface circulation of the Balearic Sea. *J. Geophys. Res.: Ocean.* 95, 1559–1568. <http://dx.doi.org/10.1029/JC095iC02p01559>.
- Lanoix, F., 1974. *Projet Alboran: Étude Hydrologique et Dynamique de La Mer d'Alboran d'après Les Travaux Effectués Dans Le Cadre Du Projet Alboran. Organisation du Traité de l'Atlantique. Sous-comité océanographique*.
- Lau, K.M., Weng, H., 1995. Climate signal detection using wavelet transform: How to make a time series sing. *Bull. Am. Meteorol. Soc.* 76, 2391–2402. [http://dx.doi.org/10.1175/1520-0477\(1995\)076<2391:CSDUWT>2.0.CO;2](http://dx.doi.org/10.1175/1520-0477(1995)076<2391:CSDUWT>2.0.CO;2).
- Le Fèvre, J., 1987. Aspects of the biology of frontal systems. *Adv. Mar. Biology* 23, [http://dx.doi.org/10.1016/S0065-2881\(08\)60109-1](http://dx.doi.org/10.1016/S0065-2881(08)60109-1).
- Lévy, M., 2003. Mesoscale variability of phytoplankton and of new production: Impact of the large-scale nutrient distribution. *J. Geophys. Res.: Ocean.* 108, <http://dx.doi.org/10.1029/2002JC001577>.
- Lévy, M., Ferrari, R., Franks, P.J.S., Martin, A.P., Rivière, P., 2012. Bringing physics to life at the submesoscale. *Geophys. Res. Lett.* 39, <http://dx.doi.org/10.1029/2012GL052756>.
- Lévy, M., Franks, P.J.S., Smith, K.S., 2018. The role of submesoscale currents in structuring marine ecosystems. *Nat. Commun.* 9, 4758. <http://dx.doi.org/10.1038/s41467-018-07059-3>.
- Lévy, M., Klein, P., Treguier, A.M., 2001. Impact of sub-mesoscale physics on production and subduction of phytoplankton in an oligotrophic regime. *J. Mar. Res.* 59, <http://dx.doi.org/10.1357/002224001762842181>.
- Longhurst, A.R., 2007. *Ecological Geography of the Sea*, second ed. Academic Press, Amsterdam; Boston, MA.
- Macías, D., Bruno, M., Echevarría, F., Vázquez, Á., García, C.M., 2008. Meteorologically-induced mesoscale variability of the North-western Alboran Sea (southern Spain) and related biological patterns. *Estuar. Coast. Shelf Sci.* 78, 250–266. <http://dx.doi.org/10.1016/j.ecss.2007.12.008>.
- Mahadevan, A., 2016. The impact of submesoscale physics on primary productivity of plankton. *Annu. Rev. Mar. Sci.* 8, 161–184. <http://dx.doi.org/10.1146/annurev-marine-010814-015912>.
- Mahadevan, A., Archer, D., 2000. Modeling the impact of fronts and mesoscale circulation on the nutrient supply and biogeochemistry of the upper ocean. *J. Geophys. Res.: Ocean.* 105, 1209–1225. <http://dx.doi.org/10.1029/1999JC000216>.
- Mahadevan, A., D'Asaro, E., Lee, C., Perry, M.J., 2012. Eddy-driven stratification initiates North Atlantic spring phytoplankton blooms. *Science* 337, 54–58. <http://dx.doi.org/10.1126/science.1218740>.
- Maritorena, S., Siegel, D.A., 2005. Consistent merging of satellite ocean color data sets using a bio-optical model. *Remote Sens. Environ.* 94, 429–440. <http://dx.doi.org/10.1016/j.rse.2004.08.014>.
- McGillicuddy, D.J., Robinson, A.R., Siegel, D.A., Jannasch, H.W., Johnson, R., Dickey, T.D., McNeil, J., Michaels, A.F., Knap, A.H., 1998. Influence of mesoscale eddies on new production in the Sargasso Sea. *Nature* 394, 263–266. <http://dx.doi.org/10.1038/28367>.
- Mena, C., Reglero, P., Ferriol, P., Torres, A.P., Aparicio-González, A., Balbín, R., Santiago, R., Moyà, G., Alemany, F., Agawin, N.S.R., 2016. Prokaryotic picoplankton spatial distribution during summer in a haline front in the Balearic Sea, Western Mediterranean. *Hydrobiologia* 779, 243–257. <http://dx.doi.org/10.1007/s10750-016-2825-4>.
- Millot, C., 1999. Circulation in the Western Mediterranean Sea. *J. Mar. Syst.* 20, 423–442. [http://dx.doi.org/10.1016/S0924-7963\(98\)00078-5](http://dx.doi.org/10.1016/S0924-7963(98)00078-5).
- Navarro, G., Vázquez, Á., Macías, D., Bruno, M., Ruiz, J., 2011. Understanding the patterns of biological response to physical forcing in the Alborán Sea (western Mediterranean). *Geophys. Res. Lett.* 38, <http://dx.doi.org/10.1029/2011GL049708>.
- Nieto, K., 2009. *Variabilidad Oceánica de Mesoscala En Los Ecosistemas de Aflojamiento de Chile y Canarias: Una Comparación a Partir de Datos Satelitales* (Ph.D. thesis). Universidad de Salamanca, Spain.
- Oguz, T., Macías, D., García-Lafuente, J., Pascual, A., Tintoré, J., 2014. Fueling plankton production by a meandering frontal jet: A case study for the Alboran Sea (Western Mediterranean). *PLoS ONE* 9, e111482. <http://dx.doi.org/10.1371/journal.pone.0111482>.
- Oguz, T., Murre, B., Tintoré, J., 2016. Upstream control of the frontal jet regulating plankton production in the Alboran Sea (Western Mediterranean): Frontal Jet regulating production. *J. Geophys. Res.: Ocean.* 121, 7159–7175. <http://dx.doi.org/10.1002/2016JC011667>.

- Olson, D.B., Hitchcock, G.L., Mariano, A.J., Ashjian, C.J., Peng, G., 2015. Life on the edge: Marine life and fronts. *Oceanography* 7, 52–60. <http://dx.doi.org/10.5670/oceanog.1994.03>.
- Omand, M.M., D'Asaro, E.A., Lee, C.M., Perry, M.J., Briggs, N., Cetinić, I., Mahadevan, A., 2015. Eddy-driven subduction exports particulate organic carbon from the spring bloom. *Science* 348, 222–225. <http://dx.doi.org/10.1126/science.1260062>.
- Oram, J.J., McWilliams, J.C., Stolzenbach, K.D., 2008. Gradient-based edge detection and feature classification of sea-surface images of the Southern California Bight. *Remote Sens. Environ.* 112, <http://dx.doi.org/10.1016/j.rse.2007.11.010>.
- Palomera, I., Olivar, M.P., Salat, J., Sabatés, A., Coll, M., García, A., Morales-Nin, B., 2007. Small pelagic fish in the NW Mediterranean Sea: An ecological review. *Prog. Oceanogr.* 74, 377–396. <http://dx.doi.org/10.1016/j.pocean.2007.04.012>.
- Parras-Berrolcal, I.M., Vazquez, R., Cabos, W., Sein, D., Mañanes, R., Perez-Sanz, J., Izquierdo, A., 2020. The climate change signal in the Mediterranean Sea in a regionally coupled atmosphere-ocean model. *Ocean. Sci.* 16, 743–765. <http://dx.doi.org/10.5194/os-16-743-2020>.
- Pascual, A., Ruiz, S., Olita, A., Troupin, C., Claret, M., Casas, B., Mourre, B., Poulain, P.M., Tovar-Sanchez, A., Capet, A., Mason, E., Allen, J.T., Mahadevan, A., Tintoré, J., 2017. A multipplatform experiment to unravel meso- and submesoscale processes in an intense front (AlborEx). *Front. Mar. Sci.* 4, <http://dx.doi.org/10.3389/fmars.2017.00039>.
- Ramírez, T., Cortés, D., Mercado, J.M., Vargas-Yañez, M., Sebastián, M., Liger, E., 2005. Seasonal dynamics of inorganic nutrients and phytoplankton biomass in the NW Alboran Sea. *Estuar. Coast. Shelf Sci.* 65, 654–670. <http://dx.doi.org/10.1016/j.ecss.2005.07.012>.
- Renault, L., Oguz, T., Pascual, A., Vizoso, G., Tintoré, J., 2012. Surface circulation in the Alborán Sea (western Mediterranean) inferred from remotely sensed data. *J. Geophys. Res.: Ocean.* 117, <http://dx.doi.org/10.1029/2011JC007659>.
- Rivière, P., Pondaven, P., 2006. Phytoplankton size classes competitions at sub-mesoscale in a frontal oceanic region. *J. Mar. Syst.* 60, 345–364. <http://dx.doi.org/10.1016/j.jmarsys.2006.02.005>.
- Rodríguez, J., Tintoré, J., Allen, J.T., Blanco, J.M., Gomis, D., Reul, A., Ruiz, J., Rodríguez, V., Echevarría, F., Jiménez-Gómez, F., 2001. Mesoscale vertical motion and the size structure of phytoplankton in the ocean. *Nature* 410, 360. <http://dx.doi.org/10.1038/35066560>.
- Rousselet, L., Doglioli, A.M., de Verneil, A., Pietri, A., Della Penna, A., Berline, L., Marrec, P., Grégori, G., Thyssen, M., Carlotti, F., Barrillon, S., Simon-Bot, F., Bonal, M., d'Ovidio, F., Petrenko, A., 2019. Vertical motions and their effects on a biogeochemical tracer in a cyclonic structure finely observed in the ligurian sea. *J. Geophys. Res.: Ocean.* 124, 3561–3574. <http://dx.doi.org/10.1029/2018JC014392>.
- Royer, F., Fromentin, J., Gaspar, P., 2004. Association between bluefin tuna schools and oceanic features in the western Mediterranean. *Mar. Ecol. Prog. Ser.* 269, 249–263. <http://dx.doi.org/10.3354/meps269249>.
- Rudnick, D.L., 1996. Intensive surveys of the Azores Front: 2. Inferring the geostrophic and vertical velocity fields. *J. Geophys. Res.: Ocean.* 101, 16291–16303. <http://dx.doi.org/10.1029/96JC01144>.
- Ruiz, S., Claret, M., Pascual, A., Olita, A., Troupin, C., Capet, A., Tovar-Sánchez, A., Allen, J., Poulain, P.M., Tintoré, J., Mahadevan, A., 2019. Effects of oceanic mesoscale and submesoscale frontal processes on the vertical transport of phytoplankton. *J. Geophys. Res.: Ocean.* 124.
- Sala, I., Bolado-Penagos, M., Bartual, A., Bruno, M., García, C.M., López-Urrutia, Á., González-García, C., Echevarría, F., 2022. A Lagrangian approach to the Atlantic Jet entering the Mediterranean Sea: Physical and biogeochemical characterization. *J. Mar. Syst.* 226, 103652. <http://dx.doi.org/10.1016/j.jmarsys.2021.103652>.
- Sarhan, T., García Lafuente, J., Vargas, M., Vargas, J.M., Plaza, F., 2000. Upwelling mechanisms in the northwestern Alboran Sea. *J. Mar. Syst.* 23, 317–331. [http://dx.doi.org/10.1016/S0924-7963\(99\)00068-8](http://dx.doi.org/10.1016/S0924-7963(99)00068-8).
- Simpson, J., Hunter, J., 1974. Fronts in the Irish Sea. *Nature* 250, 404–406. <http://dx.doi.org/10.1038/250404a0>.
- Somavilla, R., González-Pola, C., Fernández-Díaz, J., 2017. The warmer the ocean surface, the shallower the mixed layer. How much of this is true? *J. Geophys. Res.: Ocean.* 122, 7698–7716. <http://dx.doi.org/10.1002/2017JC013125>.
- Spall, S.A., Richards, K.J., 2000. A numerical model of mesoscale frontal instabilities and plankton dynamics — I. Model formulation and initial experiments. *Deep. Sea Res. Part I: Ocean. Res. Pap.* 47, 1261–1301. [http://dx.doi.org/10.1016/S0967-0637\(99\)00081-3](http://dx.doi.org/10.1016/S0967-0637(99)00081-3).
- Tintoré, J., Gomis, D., Alonso, S., Parrilla, G., 1991. Mesoscale dynamics and vertical motion in the Alborán Sea. [http://dx.doi.org/10.1175/1520-0485\(1991\)021<0811:MDAVMI>2.0.CO;2](http://dx.doi.org/10.1175/1520-0485(1991)021<0811:MDAVMI>2.0.CO;2).
- Tintoré, J., La Violette, P.E., Blade, I., Cruzado, A., 1988. A study of an intense density front in the Eastern Alboran Sea: The Almería-Orán Front. 18, pp. 1384–1397. [http://dx.doi.org/10.1175/1520-0485\(1988\)018<1384:ASOAIID>2.0.CO;2](http://dx.doi.org/10.1175/1520-0485(1988)018<1384:ASOAIID>2.0.CO;2).
- Torrence, C., Compo, G.P., 1998. A practical guide to wavelet analysis. *Bull. Am. Meteorol. Soc.* 79, 61–78. [http://dx.doi.org/10.1175/1520-0477\(1998\)079<0061:APGTWA>2.0.CO;2](http://dx.doi.org/10.1175/1520-0477(1998)079<0061:APGTWA>2.0.CO;2).
- Tzortzis, R., Doglioli, A.M., Barrillon, S., Petrenko, A.A., d'Ovidio, F., Izard, L., Thyssen, M., Pascual, A., Barceló-Llull, B., Cyr, F., Tedetti, M., Bhairy, N., Garreau, P., Dumas, F., Gregori, G., 2021. Impact of moderately energetic fine-scale dynamics on the phytoplankton community structure in the western Mediterranean Sea. *Biogeosciences* 18, 6455–6477. <http://dx.doi.org/10.5194/bg-18-6455-2021>.
- Tzortzis, R., Doglioli, A.M., Messié, M., Barrillon, S., Petrenko, A.A., Izard, L., Zhao, Y., d'Ovidio, F., Dumas, F., Gregori, G., 2023. The contrasted phytoplankton dynamics across a frontal system in the southwestern Mediterranean Sea. *Biogeosciences* 20, 3491–3508. <http://dx.doi.org/10.5194/bg-20-3491-2023>.
- Vargas-Yañez, M., Moya, F., Balbín, R., Santiago, R., Ballesteros, E., Sánchez-Leal, R.F., Romero, P., García-Martínez, M.C., 2022. Seasonal and long-term variability of the mixed layer depth and its influence on ocean productivity in the Spanish Gulf of Cádiz and Mediterranean Sea. *Front. Mar. Sci.* 9, <http://dx.doi.org/10.3389/fmars.2022.901893>.
- Vélez-Belchí, P., Vargas-Yañez, M., Tintoré, J., 2005. Observation of a western alborán gyre migration event. *Prog. Oceanogr.* 66, 190–210. <http://dx.doi.org/10.1016/j.pocean.2004.09.006>.
- Viúdez, Á., Haney, R.L., Tintoré, J., 1996. Circulation in the Alboran Sea as determined by quasi-synoptic hydrographic observations. Part II: Mesoscale ageostrophic motion diagnosed through density dynamical assimilation. *J. Phys. Oceanogr.* 26, 706–724. [http://dx.doi.org/10.1175/1520-0485\(1996\)026<0706:CITASA>2.0.CO;2](http://dx.doi.org/10.1175/1520-0485(1996)026<0706:CITASA>2.0.CO;2).
- Wang, T., Sun, Y., Su, H., Lu, W., 2023. Declined trends of chlorophyll a in the South China Sea over 2005–2019 from remote sensing reconstruction. *Acta Ocean. Sin.* 42, 12–24. <http://dx.doi.org/10.1007/s13131-022-2097-y>.
- Yang, K., Fischer, A.M., 2025. Assessing the impact of data-resolution on ocean frontal characteristics. PFG – J. Photogramm. Remote. Sens. Geoinf. Sci. 93, 113–129. <http://dx.doi.org/10.1007/s41064-024-00318-7>.
- Zarokanellos, N.D., Rudnick, D.L., Garcia-Jove, M., Mourre, B., Ruiz, S., Pascual, A., Tintoré, J., 2022. Frontal dynamics in the Alboran Sea: 1. Coherent 3D pathways at the Almería-Oran Front using underwater glider observations. *J. Geophys. Res.: Ocean.* 127, <http://dx.doi.org/10.1029/2021JC017405>.